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Method and device for forming signal field including control information

Abstract

A transmission and reception PPDU according to the present specification may include a first control signal field and a second control signal field. Overflowed information from the first control signal field may include a first encoding block of a common field of the second control signal field. In addition, the common field of the second control signal field may include various numbers of encoding blocks and various numbers of RU allocation subfields on the basis of the characteristics of the transmission and reception PPDU. In addition, the overflowed information may be included in a plurality of content channels using various methods.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2021/007457, filed on Jun. 15, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0073143, filed on Jun. 16, 2020, Korean Patent Application No. 10-2020-0104891, filed on Aug. 20, 2020 and Korean Patent Application No. 10-2020-0118618, filed on Sep. 15, 2020, which are all hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

(2) The present specification relates to a data unit in a wireless LAN, and more specifically, to a method and apparatus for configuring and decoding a field of a physical protocol data unit (PPDU) used in a wireless LAN.

BACKGROUND

(3) A wireless local area network (WLAN) has been improved in various ways. For example, the IEEE 802.11ax standard proposed an improved communication environment using orthogonal frequency division multiple access (OFDMA) and downlink multi-user multiple input multiple output (DL MU MIMO) techniques.

(4) The present specification proposes a technical feature that can be utilized in a new communication standard. For example, the new communication standard may be an extreme high throughput (EHT) standard which is currently being discussed. The EHT standard may use an increased bandwidth, an enhanced PHY layer protocol data unit (PPDU) structure, an enhanced sequence, a hybrid automatic repeat request (HARQ) scheme, or the like, which is newly proposed. The EHT standard may be called the IEEE 802.11be standard.

(5) In order to support a high throughput and a high data rate, the EHT standard may use a wide bandwidth (e.g., 160/320 MHz), 16 streams, and/or a multi-link (or multi-band) operation or the like.

(6) In the EHT standard, a wide bandwidth (e.g., 160/240/320 MHz) may be used for high throughput. Also, in order to efficiently use the bandwidth, preamble puncturing and multiple RU transmission may be used.

SUMMARY

(7) An additional control field may be configured in a new wireless LAN standard (e.g., EHT standard), and in this case, a problem in that information included in the common control field may be excessively large may occur. For example, when the number of information fields included in Universal SIG (U-SIG) according to the EHT standard increases excessively, it may be a problem how to process information overflowing from the U-SIG. Also, if the information overflowing from the U-SIG is included in another control field, how to configure the field may need to be discussed.

(8) An example of the present specification proposes an example of a control field of a PPDU according to a new wireless LAN standard. For example, an example of the present specification may propose a method/apparatus related to a TX station. According to the present specification, the TX STA may configure a transmission (TX) physical protocol data unit (PPDU) for at least one user.

(9) The TX PPDU may include a first control signal field for interpreting the TX PPDU and a second control signal field including additional control information for the TX PPD

(10) The second control signal field may be transmitted through a plurality of content channels including a first content channel and a second content channel, wherein the plurality of content channels may be allocated to different frequency bands.

(11) The second control signal field may further include a common field comprising at least one encoding block and a user specific field comprising at least one encoding block.

(12) The common field may include overflowed information from the first control signal field and at least one resource unit (RU) allocation subfield.

(13) The number of encoding blocks included in the common field transmitted through one content channel is set to two for a transmission bandwidth of the TX PPDU.

(14) An example of the present specification proposes an example of a control field of a PPDU according to a new wireless LAN standard. The second control signal field according to the present specification may include overflowed information from the first control signal field, thereby solving a technical problem caused by the limited number of bits of the first control signal field. Also, the common field of the second control signal field according to the present specification is configured based on the optimized number of encoding blocks. In addition, the common field of the second control signal field according to the present specification includes an optimized Resource Unit (RU) allocation sub-fields. Through this, individual subfields of the common field of the second control signal field can be protected from transmission/reception errors through a CRC field to be included in the second control signal field.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.

(2) FIG. 2 is a conceptual view illustrating the structure of a wireless local area network (WLAN).

(3) FIG. 3 illustrates an example of a PPDU used in an IEEE standard.

(4) FIG. 4 illustrates a layout of resource units (RUs) used in a band of 20 MHz.

(5) FIG. 5 illustrates a layout of RUs used in a band of 40 MHz.

(6) FIG. 6 illustrates a layout of RUs used in a band of 80 MHz.

(7) FIG. 7 illustrates a structure of an HE-SIG-B field.

(8) FIG. 8 illustrates an example in which a plurality of user STAs are allocated to the same RU

through a MU-MIMO scheme.

(9) FIG. 9 illustrates an operation based on UL-MU.

(10) FIG. 10 illustrates an example of a channel used/supported/defined within a 2.4 GHz band.

(11) FIG. 11 illustrates an example of a channel used/supported/defined within a 5 GHz band.

(12) FIG. 12 illustrates an example of a channel used/supported/defined within a 6 GHz band.

(13) FIG. 13 illustrates an example of a PPDU used in the present specification.

(14) FIG. 14 illustrates an example of a modified transmitting apparatus and/or receiving apparatus of the present specification.

(15) FIG. 15 shows an example of an aggregation of RU26 and RU52 in 20 MHz.

(16) FIG. 16 shows an example of an aggregation of RU26 and RU52 in 40 MHz.

(17) FIG. 17 shows an example of an aggregation of RU26 and RU52 in 80 MHz.

(18) FIG. 18 shows an example of an EHT PPDU.

(19) FIG. 19 shows an example of a first control signal field or U-SIG field of the present specification.

(20) FIG. 20 shows an example in which the overflowed information of this specification is repeated in two content channels.

(21) FIG. 21 shows a common field of a control signal field according to an example of the present specification.

(22) FIG. 22 shows an additional example of a common field of a control signal field according to an example of the present specification.

(23) FIG. 23 shows an additional example of a common field of a control signal field according to an example of the present specification.

(24) FIG. 24 shows an additional example of a common field of a control signal field according to an example of the present specification.

(25) FIG. 25 is a flow chart illustrating operations performed by a TX STA.

(26) FIG. 26 is another flow chart illustrating operations performed by a TX STA.

(27) FIG. 27 is a flow chart illustrating operations performed by a RX STA.

(28) FIG. 28 is an example of a content channel for a 20/40/80 MHz PPDU.

(29) FIG. 29 is an example of a content channel for a 160 MHz PPDU.

(30) FIG. 30 is an example of a content channel for a 320 MHz PPDU.

(31) FIG. 31 is an example of a common field and a user specific field of the present specification.

DETAILED DESCRIPTION

(32) In the present specification, “A or B” may mean “only A”, “only B” or “both A and B”. In other words, in the present specification, “A or B” may be interpreted as “A and/or B”. For example, in the present specification, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.

(33) A slash (/) or comma used in the present specification may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.

(34) In the present specification, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present specification, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.

(35) In addition, in the present specification, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, and C”. In addition, “at least one of A, B, or C” or “at least one of A, B, and/or C” may mean “at least one of A, B, and C”.

(36) In addition, a parenthesis used in the present specification may mean “for example”.

Specifically, when indicated as “control information (EHT-signal)”, it may mean that “EHT-signal” is proposed as an example of the “control information”. In other words, the “control information” of the present specification is not limited to “EHT-signal”, and “EHT-signal” may be proposed as an example of the “control information”. In addition, when indicated as “control information (i.e.,

EHT-signal)", it may also mean that "EHT-signal" is proposed as an example of the "control information".

(37) Technical features described individually in one figure in the present specification may be individually implemented, or may be simultaneously implemented.

(38) The following example of the present specification may be applied to various wireless communication systems. For example, the following example of the present specification may be applied to a wireless local area network (WLAN) system. For example, the present specification may be applied to the IEEE 802.11a/g/n/ac standard or the IEEE 802.11ax standard. In addition, the present specification may also be applied to the newly proposed EHT standard or IEEE 802.11be standard. In addition, the example of the present specification may also be applied to a new WLAN standard enhanced from the EHT standard or the IEEE 802.11be standard. In addition, the example of the present specification may be applied to a mobile communication system. For example, it may be applied to a mobile communication system based on long term evolution (LTE) depending on a 3rd generation partnership project (3GPP) standard and based on evolution of the LTE. In addition, the example of the present specification may be applied to a communication system of a 5G NR standard based on the 3GPP standard.

(39) Hereinafter, in order to describe a technical feature of the present specification, a technical feature applicable to the present specification will be described.

(40) FIG. 1 shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.

(41) In the example of FIG. 1, various technical features described below may be performed. FIG. 1 relates to at least one station (STA). For example, STAs **110** and **120** of the present specification may also be called in various terms such as a mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, or simply a user. The STAs **110** and **120** of the present specification may also be called in various terms such as a network, a base station, a node-B, an access point (AP), a repeater, a router, a relay, or the like. The STAs **110** and **120** of the present specification may also be referred to as various names such as a receiving apparatus, a transmitting apparatus, a receiving STA, a transmitting STA, a receiving device, a transmitting device, or the like.

(42) For example, the STAs **110** and **120** may serve as an AP or a non-AP. That is, the STAs **110** and **120** of the present specification may serve as the AP and/or the non-AP. In the present specification, the AP may be indicated as an AP STA.

(43) The STAs **110** and **120** of the present specification may support various communication standards together in addition to the IEEE 802.11 standard. For example, a communication standard (e.g., LTE, LTE-A, 5G NR standard) or the like based on the 3GPP standard may be supported. In addition, the STA of the present specification may be implemented as various devices such as a mobile phone, a vehicle, a personal computer, or the like. In addition, the STA of the present specification may support communication for various communication services such as voice calls, video calls, data communication, and self-driving (autonomous-driving), or the like.

(44) The STAs **110** and **120** of the present specification may include a medium access control (MAC) conforming to the IEEE 802.11 standard and a physical layer interface for a radio medium.

(45) The STAs **110** and **120** will be described below with reference to a sub-figure (a) of FIG. 1.

(46) The first STA **110** may include a processor **111**, a memory **112**, and a transceiver **113**. The illustrated process, memory, and transceiver may be implemented individually as separate chips, or at least two blocks/functions may be implemented through a single chip.

(47) The transceiver **113** of the first STA performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be, etc.) may be transmitted/received.

(48) For example, the first STA **110** may perform an operation intended by an AP. For example, the processor **111** of the AP may receive a signal through the transceiver **113**, process a reception (RX)

signal, generate a transmission (TX) signal, and provide control for signal transmission. The memory **112** of the AP may store a signal (e.g., RX signal) received through the transceiver **113**, and may store a signal (e.g., TX signal) to be transmitted through the transceiver.

(49) For example, the second STA **120** may perform an operation intended by a non-AP STA. For example, a transceiver **123** of a non-AP performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be packet, etc.) may be transmitted/received.

(50) For example, a processor **121** of the non-AP STA may receive a signal through the transceiver **123**, process an RX signal, generate a TX signal, and provide control for signal transmission. A memory **122** of the non-AP STA may store a signal (e.g., RX signal) received through the transceiver **123**, and may store a signal (e.g., TX signal) to be transmitted through the transceiver.

(51) For example, an operation of a device indicated as an AP in the specification described below may be performed in the first STA **110** or the second STA **120**. For example, if the first STA **110** is the AP, the operation of the device indicated as the AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **112** of the first STA **110**. In addition, if the second STA **120** is the AP, the operation of the device indicated as the AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **122** of the second STA **120**.

(52) For example, in the specification described below, an operation of a device indicated as a non-AP (or user-STA) may be performed in the first STA **110** or the second STA **120**. For example, if the second STA **120** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **122** of the second STA **120**. For example, if the first STA **110** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **112** of the first STA **110**.

(53) In the specification described below, a device called a (transmitting/receiving) STA, a first STA, a second STA, an STA1, an STA2, an AP, a first AP, a second AP, an AP1, an AP2, a (transmitting/receiving) terminal, a (transmitting/receiving) device, a (transmitting/receiving) apparatus, a network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, a device indicated as, without a specific reference numeral, the (transmitting/receiving) STA, the first STA, the second STA, the STA1, the STA2, the AP, the first AP, the second AP, the AP1, the AP2, the (transmitting/receiving) terminal, the (transmitting/receiving) device, the (transmitting/receiving) apparatus, the network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, in the following example, an operation in which various STAs transmit/receive a signal (e.g., a PPDU) may be performed in the transceivers **113** and **123** of FIG. **1**. In addition, in the following example, an operation in which various STAs generate a TX/RX signal or perform data processing and computation in advance for the TX/RX signal may be performed in the processors **111** and **121** of FIG. **1**. For example, an example of an operation for generating the TX/RX signal or performing the data processing and computation in advance may include: 1) an operation of determining/obtaining/configuring/computing/decoding/encoding bit information of a sub-field (SIG, STF, LTF, Data) included in a PPDU; 2) an operation of determining/configuring/obtaining a

time resource or frequency resource (e.g., a subcarrier resource) or the like used for the sub-field (SIG, STF, LTF, Data) included in the PPDU; 3) an operation of determining/configuring/obtaining a specific sequence (e.g., a pilot sequence, an STF/LTF sequence, an extra sequence applied to SIG) or the like used for the sub-field (SIG, STF, LTF, Data) field included in the PPDU; 4) a power control operation and/or power saving operation applied for the STA; and 5) an operation related to determining/obtaining/configuring/decoding/encoding or the like of an ACK signal. In addition, in the following example, a variety of information used by various STAs for determining/obtaining/configuring/decoding/encoding a TX/RX signal (e.g., information related to a field/subfield/control field/parameter/power or the like) may be stored in the memories **112** and **122** of FIG. **1**.

(54) The aforementioned device/STA of the sub-figure (a) of FIG. **1** may be modified as shown in the sub-figure (b) of FIG. **1**. Hereinafter, the STAs **110** and **120** of the present specification will be described based on the sub-figure (b) of FIG. **1**.

(55) For example, the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned transceiver illustrated in the sub-figure (a) of FIG. **1**. For example, processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1** may include the processors **111** and **121** and the memories **112** and **122**. The processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (a) of FIG. **1**.

(56) A mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, a user, a user STA, a network, a base station, a Node-B, an access point (AP), a repeater, a router, a relay, a receiving unit, a transmitting unit, a receiving STA, a transmitting STA, a receiving device, a transmitting device, a receiving apparatus, and/or a transmitting apparatus, which are described below, may imply the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may imply the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. That is, a technical feature of the present specification may be performed in the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may be performed only in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. For example, a technical feature in which the transmitting STA transmits a control signal may be understood as a technical feature in which a control signal generated in the processors **111** and **121** illustrated in the sub-figure (a)/(b) of FIG. **1** is transmitted through the transceivers **113** and **123** illustrated in the sub-figure (a)/(b) of FIG. **1**. Alternatively, the technical feature in which the transmitting STA transmits the control signal may be understood as a technical feature in which the control signal to be transferred to the transceivers **113** and **123** is generated in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(57) For example, a technical feature in which the receiving STA receives the control signal may be understood as a technical feature in which the control signal is received by means of the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1** is obtained by the processors **111** and **121** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** is obtained by the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(58) Referring to the sub-figure (b) of FIG. **1**, software codes **115** and **125** may be included in the memories **112** and **122**. The software codes **115** and **126** may include instructions for controlling an operation of the processors **111** and **121**. The software codes **115** and **125** may be included as various programming languages.

(59) The processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include an application-specific integrated circuit (ASIC), other chipsets, a logic circuit and/or a data processing device. The processor may be an application processor (AP). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include at least one of a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), and a modulator and demodulator (modem). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may be SNAPDRAGON™ series of processors made by Qualcomm®, EXYNOS™ series of processors made by Samsung®, A series of processors made by Apple®, HELIO™ series of processors made by MediaTek®, ATOM™ series of processors made by Intel® or processors enhanced from these processors.

(60) In the present specification, an uplink may imply a link for communication from a non-AP STA to an SP STA, and an uplink PPDU/packet/signal or the like may be transmitted through the uplink. In addition, in the present specification, a downlink may imply a link for communication from the AP STA to the non-AP STA, and a downlink PPDU/packet/signal or the like may be transmitted through the downlink.

(61) FIG. **2** is a conceptual view illustrating the structure of a wireless local area network (WLAN).

(62) An upper part of FIG. **2** illustrates the structure of an infrastructure basic service set (BSS) of institute of electrical and electronic engineers (IEEE) 802.11.

(63) Referring the upper part of FIG. **2**, the wireless LAN system may include one or more infrastructure BSSs **200** and **205** (hereinafter, referred to as BSS). The BSSs **200** and **205** as a set of an AP and an STA such as an access point (AP) **225** and a station (STA1) **200-1** which are successfully synchronized to communicate with each other are not concepts indicating a specific region. The BSS **205** may include one or more STAs **205-1** and **205-2** which may be joined to one AP **230**.

(64) The BSS may include at least one STA, APs providing a distribution service, and a distribution system (DS) **210** connecting multiple APs.

(65) The distribution system **210** may implement an extended service set (ESS) **240** extended by connecting the multiple BSSs **200** and **205**. The ESS **240** may be used as a term indicating one network configured by connecting one or more APs **225** or **230** through the distribution system **210**. The AP included in one ESS **240** may have the same service set identification (SSID).

(66) A portal **220** may serve as a bridge which connects the wireless LAN network (IEEE 802.11) and another network (e.g., 802.X).

(67) In the BSS illustrated in the upper part of FIG. **2**, a network between the APs **225** and **230** and a network between the APs **225** and **230** and the STAs **200-1**, **205-1**, and **205-2** may be implemented. However, the network is configured even between the STAs without the APs **225** and **230** to perform communication. A network in which the communication is performed by configuring the network even between the STAs without the APs **225** and **230** is defined as an Ad-Hoc network or an independent basic service set (IBSS).

(68) A lower part of FIG. **2** illustrates a conceptual view illustrating the IBSS.

(69) Referring to the lower part of FIG. **2**, the IBSS is a BSS that operates in an Ad-Hoc mode. Since the IBSS does not include the access point (AP), a centralized management entity that performs a management function at the center does not exist. That is, in the IBSS, STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** are managed by a distributed manner. In the IBSS, all STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** may be constituted by movable STAs and are not permitted to access the DS to constitute a self-contained network.

(70) FIG. **3** illustrates an example of a PPDU used in an IEEE standard.

(71) As illustrated in FIG. **3**, various types of PHY protocol data units (PPDUs) are used in IEEE a/g/n/ac standards. Specifically, an LTF and a STF include a training signal, a SIG-A and a SIG-B include control information for a receiving STA, and a data field includes user data corresponding to a PSDU (MAC PDU/aggregated MAC PDU).

(72) FIG. 3 also includes an example of an HE PPDU according to IEEE 802.11ax. The HE PPDU according to FIG. 3 is an illustrative PPDU for multiple users. An HE-SIG-B may be included only in a PPDU for multiple users, and an HE-SIG-B may be omitted in a PPDU for a single user.

(73) As illustrated in FIG. 3, the HE-PPDU for multiple users (MUs) may include a legacy-short training field (L-STF), a legacy-long training field (L-LTF), a legacy-signal (L-SIG), a high efficiency-signal A (HE-SIG A), a high efficiency-signal-B (HE-SIG B), a high efficiency-short training field (HE-STF), a high efficiency-long training field (HE-LTF), a data field (alternatively, an MAC payload), and a packet extension (PE) field. The respective fields may be transmitted for illustrated time periods (i.e., 4 or 8 μ s).

(74) Hereinafter, a resource unit (RU) used for a PPDU is described. An RU may include a plurality of subcarriers (or tones). An RU may be used to transmit a signal to a plurality of STAs according to OFDMA. Further, an RU may also be defined to transmit a signal to one STA. An RU may be used for an STF, an LTF, a data field, or the like.

(75) FIG. 4 illustrates a layout of resource units (RUs) used in a band of 20 MHz.

(76) As illustrated in FIG. 4, resource units (RUs) corresponding to different numbers of tones (i.e., subcarriers) may be used to form some fields of an HE-PPDU. For example, resources may be allocated in illustrated RUs for an HE-STF, an HE-LTF, and a data field.

(77) As illustrated in the uppermost part of FIG. 4, a 26-unit (i.e., a unit corresponding to 26 tones) may be disposed. Six tones may be used for a guard band in the leftmost band of the 20 MHz band, and five tones may be used for a guard band in the rightmost band of the 20 MHz band. Further, seven DC tones may be inserted in a center band, that is, a DC band, and a 26-unit corresponding to 13 tones on each of the left and right sides of the DC band may be disposed. A 26-unit, a 52-unit, and a 106-unit may be allocated to other bands. Each unit may be allocated for a receiving STA, that is, a user.

(78) The layout of the RUs in FIG. 4 may be used not only for a multiple users (MUs) but also for a single user (SU), in which case one 242-unit may be used and three DC tones may be inserted as illustrated in the lowermost part of FIG. 4.

(79) Although FIG. 4 proposes RUs having various sizes, that is, a 26-RU, a 52-RU, a 106-RU, and a 242-RU, specific sizes of RUs may be extended or increased. Therefore, the present embodiment is not limited to the specific size of each RU (i.e., the number of corresponding tones).

(80) FIG. 5 illustrates a layout of RUs used in a band of 40 MHz.

(81) Similar to FIG. 4 in which RUs having various sizes are used, a 26-RU, a 52-RU, a 106-RU, a 242-RU, a 484-RU, and the like may be used in an example of FIG. 5. Further, five DC tones may be inserted in a center frequency, 12 tones may be used for a guard band in the leftmost band of the 40 MHz band, and 11 tones may be used for a guard band in the rightmost band of the 40 MHz band.

(82) As illustrated, when the layout of the RUs is used for a single user, a 484-RU may be used. The specific number of RUs may be changed similar to FIG. 5.

(83) FIG. 6 illustrates a layout of RUs used in a band of 80 MHz.

(84) Similar to FIG. 4 and FIG. 5 in which RUs having various sizes are used, a 26-RU, a 52-RU, a 106-RU, a 242-RU, a 484-RU, a 996-RU, and the like may be used in an example of FIG. 6. Further, seven DC tones may be inserted in the center frequency, 12 tones may be used for a guard band in the leftmost band of the 80 MHz band, and 11 tones may be used for a guard band in the rightmost band of the 80 MHz band. In addition, a 26-RU corresponding to 13 tones on each of the left and right sides of the DC band may be used.

(85) As illustrated, when the layout of the RUs is used for a single user, a 996-RU may be used, in which case five DC tones may be inserted.

(86) The RU described in the present specification may be used in uplink (UL) communication and downlink (DL) communication. For example, when UL-MU communication which is solicited by a trigger frame is performed, a transmitting STA (e.g., AP) may allocate a first RU (e.g.,

26/52/106/242-RU, etc.) to a first STA through the trigger frame, and may allocate a second RU (e.g., 26/52/106/242-RU, etc.) to a second STA. Thereafter, the first STA may transmit a first trigger-based PPDU based on the first RU, and the second STA may transmit a second trigger-based PPDU based on the second RU. The first/second trigger-based PPDU is transmitted to the AP at the same (or overlapped) time period.

(87) For example, when a DL MU PPDU is configured, the transmitting STA (e.g., AP) may allocate the first RU (e.g., 26/52/106/242-RU, etc.) to the first STA, and may allocate the second RU (e.g., 26/52/106/242-RU, etc.) to the second STA. That is, the transmitting STA (e.g., AP) may transmit HE-STF, HE-LTF, and Data fields for the first STA through the first RU in one MU PPDU, and may transmit HE-STF, HE-LTF, and Data fields for the second STA through the second RU.

(88) Information related to a layout of the RU may be signaled through HE-SIG-B.

(89) FIG. 7 illustrates a structure of an HE-SIG-B field.

(90) As illustrated, an HE-SIG-B field **710** includes a common field **720** and a user-specific field **730**. The common field **720** may include information commonly applied to all users (i.e., user STAs) which receive SIG-B. The user-specific field **730** may be called a user-specific control field. When the SIG-B is transferred to a plurality of users, the user-specific field **730** may be applied only any one of the plurality of users.

(91) As illustrated, the common field **720** and the user-specific field **730** may be separately encoded.

(92) The common field **720** may include RU allocation information of $N \times 8$ bits. For example, the RU allocation information may include information related to a location of an RU. For example, when a 20 MHz channel is used as shown in FIG. 4, the RU allocation information may include information related to a specific frequency band to which a specific RU (26-RU/52-RU/106-RU) is arranged.

(93) An example of a case in which the RU allocation information consists of 8 bits is as follows.

(94) TABLE-US-00001 TABLE 1 8 bits indices (B7 B6 B5 B4 Number B3 B2 B1 B0) #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 00000000 26 26 26 26 26 26 26 26 1 00000001 26 26 26 26 26 26 26 26 52 1 00000010 26 26 26 26 26 52 26 26 1 00000011 26 26 26 26 26 52 52 1 00000100 26 26 52 26 26 26 26 1 00000101 26 26 52 26 26 26 52 1 00000110 26 26 52 26 52 26 26 1 00000111 26 26 52 26 52 52 1 00001000 52 26 26 26 26 26 26 1

(95) As shown the example of FIG. 4, up to nine 26-RUs may be allocated to the 20 MHz channel. When the RU allocation information of the common field **720** is set to “00000000” as shown in Table 1, the nine 26-RUs may be allocated to a corresponding channel (i.e., 20 MHz). In addition, when the RU allocation information of the common field **720** is set to “00000001” as shown in Table 1, seven 26-RUs and one 52-RU are arranged in a corresponding channel. That is, in the example of FIG. 4, the 52-RU may be allocated to the rightmost side, and the seven 26-RUs may be allocated to the left thereof.

(96) The example of Table 1 shows only some of RU locations capable of displaying the RU allocation information.

(97) For example, the RU allocation information may include an example of Table 2 below.

(98) TABLE-US-00002 TABLE 2 8 bits indices (B7 B6 B5 B4 Number B3 B2 B1 B0) #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 01000y.sub.2y.sub.1y.sub.0 106 26 26 26 26 26 8 01001y.sub.2y.sub.1y.sub.0 106 26 26 26 52 8

(99) “01000y2y1y0” relates to an example in which a 106-RU is allocated to the leftmost side of the 20 MHz channel, and five 26-RUs are allocated to the right side thereof. In this case, a plurality of STAs (e.g., user-STAs) may be allocated to the 106-RU, based on a MU-MIMO scheme. Specifically, up to 8 STAs (e.g., user-STAs) may be allocated to the 106-RU, and the number of STAs (e.g., user-STAs) allocated to the 106-RU is determined based on 3-bit information (y2y1y0). For example, when the 3-bit information (y2y1y0) is set to N, the number of STAs (e.g., user-STAs) allocated to the 106-RU based on the MU-MIMO scheme may be $N+1$.

based on the MU-MIMO scheme as shown in FIG. 8, N_user is set to "3". Therefore, values of N_STS[1], N_STS[2], and N_STS[3] may be determined as shown in Table 3. For example, when a value of the second bit (B11-B14) is "0011", it may be set to N_STS[1]=4, N_STS[2]=1, N_STS[3]=1. That is, in the example of FIG. 8, four spatial streams may be allocated to the user field 1, one spatial stream may be allocated to the user field 1, and one spatial stream may be allocated to the user field 3.

(112) As shown in the example of Table 3 and/or Table 4, information (i.e., the second bit, B11-B14) related to the number of spatial streams for the user STA may consist of 4 bits. In addition, the information (i.e., the second bit, B11-B14) on the number of spatial streams for the user STA may support up to eight spatial streams. In addition, the information (i.e., the second bit, B11-B14) on the number of spatial streams for the user STA may support up to four spatial streams for one user STA.

(113) In addition, a third bit (i.e., B15-18) in the user field (i.e., 21 bits) may include modulation and coding scheme (MCS) information. The MCS information may be applied to a data field in a PPDU including corresponding SIG-B.

(114) An MCS, MCS information, an MCS index, an MCS field, or the like used in the present specification may be indicated by an index value. For example, the MCS information may be indicated by an index 0 to an index 11. The MCS information may include information related to a constellation modulation type (e.g., BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, etc.) and information related to a coding rate (e.g., 1/2, 2/3, 3/4, 5/6e, etc.). Information related to a channel coding type (e.g., LCC or LDPC) may be excluded in the MCS information.

(115) In addition, a fourth bit (i.e., B19) in the user field (i.e., 21 bits) may be a reserved field.

(116) In addition, a fifth bit (i.e., B20) in the user field (i.e., 21 bits) may include information related to a coding type (e.g., BCC or LDPC). That is, the fifth bit (i.e., B20) may include information related to a type (e.g., BCC or LDPC) of channel coding applied to the data field in the PPDU including the corresponding SIG-B.

(117) The aforementioned example relates to the user field of the first format (the format of the MU-MIMO scheme). An example of the user field of the second format (the format of the non-MU-MIMO scheme) is as follows.

(118) A first bit (e.g., B0-B10) in the user field of the second format may include identification information of a user STA. In addition, a second bit (e.g., B11-B13) in the user field of the second format may include information related to the number of spatial streams applied to a corresponding RU. In addition, a third bit (e.g., B14) in the user field of the second format may include information related to whether a beamforming steering matrix is applied. A fourth bit (e.g., B15-B18) in the user field of the second format may include modulation and coding scheme (MCS) information. In addition, a fifth bit (e.g., B19) in the user field of the second format may include information related to whether dual carrier modulation (DCM) is applied. In addition, a sixth bit (i.e., B20) in the user field of the second format may include information related to a coding type (e.g., BCC or LDPC).

(119) FIG. 9 illustrates an operation based on UL-MU. As illustrated, a transmitting STA (e.g., AP) may perform channel access through contending (e.g., a backoff operation), and may transmit a trigger frame 930. That is, the transmitting STA may transmit a PPDU including the trigger frame 930. Upon receiving the PPDU including the trigger frame, a trigger-based (TB) PPDU is transmitted after a delay corresponding to SIFS.

(120) TB PDUs 941 and 942 may be transmitted at the same time period, and may be transmitted from a plurality of STAs (e.g., user STAs) having AIDs indicated in the trigger frame 930. An ACK frame 950 for the TB PPDU may be implemented in various forms.

(121) FIG. 10 illustrates an example of a channel used/supported/defined within a 2.4 GHz band.

(122) The 2.4 GHz band may be called in other terms such as a first band. In addition, the 2.4 GHz band may imply a frequency domain in which channels of which a center frequency is close to 2.4

GHz (e.g., channels of which a center frequency is located within 2.4 to 2.5 GHz) are used/supported/defined.

(123) A plurality of 20 MHz channels may be included in the 2.4 GHz band. 20 MHz within the 2.4 GHz may have a plurality of channel indices (e.g., an index 1 to an index 14). For example, a center frequency of a 20 MHz channel to which a channel index 1 is allocated may be 2.412 GHz, a center frequency of a 20 MHz channel to which a channel index 2 is allocated may be 2.417 GHz, and a center frequency of a 20 MHz channel to which a channel index N is allocated may be $(2.407+0.005*N)$ GHz. The channel index may be called in various terms such as a channel number or the like. Specific numerical values of the channel index and center frequency may be changed.

(124) FIG. 10 exemplifies 4 channels within a 2.4 GHz band. Each of 1st to 4th frequency domains **1010** to **1040** shown herein may include one channel. For example, the 1st frequency domain **1010** may include a channel 1 (a 20 MHz channel having an index 1). In this case, a center frequency of the channel 1 may be set to 2412 MHz. The 2nd frequency domain **1020** may include a channel 6. In this case, a center frequency of the channel 6 may be set to 2437 MHz. The 3rd frequency domain **1030** may include a channel 11. In this case, a center frequency of the channel 11 may be set to 2462 MHz. The 4th frequency domain **1040** may include a channel 14. In this case, a center frequency of the channel 14 may be set to 2484 MHz.

(125) FIG. 11 illustrates an example of a channel used/supported/defined within a 5 GHz band.

(126) The 5 GHz band may be called in other terms such as a second band or the like. The 5 GHz band may imply a frequency domain in which channels of which a center frequency is greater than or equal to 5 GHz and less than 6 GHz (or less than 5.9 GHz) are used/supported/defined.

Alternatively, the 5 GHz band may include a plurality of channels between 4.5 GHz and 5.5 GHz. A specific numerical value shown in FIG. 11 may be changed.

(127) A plurality of channels within the 5 GHz band include an unlicensed national information infrastructure (UNII)-1, a UNII-2, a UNII-3, and an ISM. The UNII-1 may be called UNII Low. The UNII-2 may include a frequency domain called UNII Mid and UNII-2Extended. The UNII-3 may be called UNII-Upper.

(128) A plurality of channels may be configured within the 5 GHz band, and a bandwidth of each channel may be variously set to, for example, 20 MHz, 40 MHz, 80 MHz, 160 MHz, or the like. For example, 5170 MHz to 5330 MHz frequency domains/ranges within the UNII-1 and UNII-2 may be divided into eight 20 MHz channels. The 5170 MHz to 5330 MHz frequency domains/ranges may be divided into four channels through a 40 MHz frequency domain. The 5170 MHz to 5330 MHz frequency domains/ranges may be divided into two channels through an 80 MHz frequency domain. Alternatively, the 5170 MHz to 5330 MHz frequency domains/ranges may be divided into one channel through a 160 MHz frequency domain.

(129) FIG. 12 illustrates an example of a channel used/supported/defined within a 6 GHz band.

(130) The 6 GHz band may be called in other terms such as a third band or the like. The 6 GHz band may imply a frequency domain in which channels of which a center frequency is greater than or equal to 5.9 GHz are used/supported/defined. A specific numerical value shown in FIG. 12 may be changed.

(131) For example, the 20 MHz channel of FIG. 12 may be defined starting from 5.940 GHz. Specifically, among 20 MHz channels of FIG. 12, the leftmost channel may have an index 1 (or a channel index, a channel number, etc.), and 5.945 GHz may be assigned as a center frequency. That is, a center frequency of a channel of an index N may be determined as $(5.940+0.005*N)$ GHz.

(132) Accordingly, an index (or channel number) of the 20 MHz channel of FIG. 12 may be 1, 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61, 65, 69, 73, 77, 81, 85, 89, 93, 97, 101, 105, 109, 113, 117, 121, 125, 129, 133, 137, 141, 145, 149, 153, 157, 161, 165, 169, 173, 177, 181, 185, 189, 193, 197, 201, 205, 209, 213, 217, 221, 225, 229, 233. In addition, according to the aforementioned $(5.940+0.005*N)$ GHz rule, an index of the 40 MHz channel of FIG. 13 may be 3, 11, 19, 27, 35,

43, 51, 59, 67, 75, 83, 91, 99, 107, 115, 123, 131, 139, 147, 155, 163, 171, 179, 187, 195, 203, 211, 219, 227.

(133) Although 20, 40, 80, and 160 MHz channels are illustrated in the example of FIG. 12, a 240 MHz channel or a 320 MHz channel may be additionally added.

(134) Hereinafter, a PPDU transmitted/received in an STA of the present specification will be described.

(135) FIG. 13 illustrates an example of a PPDU used in the present specification.

(136) The PPDU of FIG. 13 may be called in various terms such as an EHT PPDU, a TX PPDU, an RX PPDU, a first type or N-th type PPDU, or the like. For example, in the present specification, the PPDU or the EHT PPDU may be called in various terms such as a TX PPDU, a RX PPDU, a first type or N-th type PPDU, or the like. In addition, the EHT PPDU may be used in an EHT system and/or a new WLAN system enhanced from the EHT system.

(137) The PPDU of FIG. 13 may indicate the entirety or part of a PPDU type used in the EHT system. For example, the example of FIG. 13 may be used for both of a single-user (SU) mode and a multi-user (MU) mode. In other words, the PPDU of FIG. 13 may be a PPDU for one receiving STA or a plurality of receiving STAs. When the PPDU of FIG. 14 is used for a trigger-based (TB) mode, the EHT-SIG of FIG. 13 may be omitted. In other words, an STA which has received a trigger frame for uplink-MU (UL-MU) may transmit the PPDU in which the EHT-SIG is omitted in the example of FIG. 13.

(138) In FIG. 13, an L-STF to an EHT-LTF may be called a preamble or a physical preamble, and may be generated/transmitted/received/obtained/decoded in a physical layer.

(139) A subcarrier spacing of the L-STF, L-LTF, L-SIG, RL-SIG, U-SIG, and EHT-SIG fields of FIG. 13 may be determined as 312.5 kHz, and a subcarrier spacing of the EHT-STF, EHT-LTF, and Data fields may be determined as 78.125 kHz. That is, a tone index (or subcarrier index) of the L-STF, L-LTF, L-SIG, RL-SIG, U-SIG, and EHT-SIG fields may be expressed in unit of 312.5 kHz, and a tone index (or subcarrier index) of the EHT-STF, EHT-LTF, and Data fields may be expressed in unit of 78.125 kHz.

(140) In the PPDU of FIG. 13, the L-LTF and the L-STF may be the same as those in the conventional fields.

(141) The L-SIG field of FIG. 13 may include, for example, bit information of 24 bits. For example, the 24-bit information may include a rate field of 4 bits, a reserved bit of 1 bit, a length field of 12 bits, a parity bit of 1 bit, and a tail bit of 6 bits. For example, the length field of 12 bits may include information related to a length or time duration of a PPDU. For example, the length field of 12 bits may be determined based on a type of the PPDU. For example, when the PPDU is a non-HT, HT, VHT PPDU or an EHT PPDU, a value of the length field may be determined as a multiple of 3. For example, when the PPDU is an HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2. In other words, for the non-HT, HT, VHT PPDU or the EHT PPDU, the value of the length field may be determined as a multiple of 3, and for the HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2.

(142) For example, the transmitting STA may apply BCC encoding based on a 1/2 coding rate to the 24-bit information of the L-SIG field. Thereafter, the transmitting STA may obtain a BCC coding bit of 48 bits. BPSK modulation may be applied to the 48-bit coding bit, thereby generating 48 BPSK symbols. The transmitting STA may map the 48 BPSK symbols to positions except for a pilot subcarrier{subcarrier index -21, -7, +7, +21} and a DC subcarrier{subcarrier index 0}. As a result, the 48 BPSK symbols may be mapped to subcarrier indices -26 to -22, -20 to -8, -6 to -1, +1 to +6, +8 to +20, and +22 to +26. The transmitting STA may additionally map a signal of {-1, -1, -1, 1} to a subcarrier index{-28, -27, +27, +28}. The aforementioned signal may be used for channel estimation on a frequency domain corresponding to {-28, -27, +27, +28}.

(143) The transmitting STA may generate an RL-SIG generated in the same manner as the L-SIG.

BPSK modulation may be applied to the RL-SIG. The receiving STA may know that the RX PPDU is the HE PPDU or the EHT PPDU, based on the presence of the RL-SIG.

(144) A universal SIG (U-SIG) may be inserted after the RL-SIG of FIG. 13. The U-SIG may be called in various terms such as a first SIG field, a first SIG, a first type SIG, a control signal, a control signal field, a first (type) control signal, or the like.

(145) The U-SIG may include information of N bits, and may include information for identifying a type of the EHT PPDU. For example, the U-SIG may be configured based on two symbols (e.g., two contiguous OFDM symbols). Each symbol (e.g., OFDM symbol) for the U-SIG may have a duration of 4 us. Each symbol of the U-SIG may be used to transmit the 26-bit information. For example, each symbol of the U-SIG may be transmitted/received based on 52 data tones and 4 pilot tones.

(146) Through the U-SIG (or U-SIG field), for example, A-bit information (e.g., 52 un-coded bits) may be transmitted. A first symbol of the U-SIG may transmit first X-bit information (e.g., 26 un-coded bits) of the A-bit information, and a second symbol of the U-SIG may transmit the remaining Y-bit information (e.g. 26 un-coded bits) of the A-bit information. For example, the transmitting STA may obtain 26 un-coded bits included in each U-SIG symbol. The transmitting STA may perform convolutional encoding (i.e., BCC encoding) based on a rate of $R=1/2$ to generate 52-coded bits, and may perform interleaving on the 52-coded bits. The transmitting STA may perform BPSK modulation on the interleaved 52-coded bits to generate 52 BPSK symbols to be allocated to each U-SIG symbol. One U-SIG symbol may be transmitted based on 65 tones (subcarriers) from a subcarrier index -28 to a subcarrier index +28, except for a DC index 0. The 52 BPSK symbols generated by the transmitting STA may be transmitted based on the remaining tones (subcarriers) except for pilot tones, i.e., tones -21, -7, +7, +21.

(147) For example, the A-bit information (e.g., 52 un-coded bits) generated by the U-SIG may include a CRC field (e.g., a field having a length of 4 bits) and a tail field (e.g., a field having a length of 6 bits). The CRC field and the tail field may be transmitted through the second symbol of the U-SIG. The CRC field may be generated based on 26 bits allocated to the first symbol of the U-SIG and the remaining 16 bits except for the CRC/tail fields in the second symbol, and may be generated based on the conventional CRC calculation algorithm. In addition, the tail field may be used to terminate trellis of a convolutional decoder, and may be set to, for example, '000000'.

(148) The A-bit information (e.g., 52 un-coded bits) transmitted by the U-SIG (or U-SIG field) may be divided into version-independent bits and version-dependent bits. For example, the version-independent bits may have a fixed or variable size. For example, the version-independent bits may be allocated only to the first symbol of the U-SIG, or the version-independent bits may be allocated to both of the first and second symbols of the U-SIG. For example, the version-independent bits and the version-dependent bits may be called in various terms such as a first control bit, a second control bit, or the like.

(149) For example, the version-independent bits of the U-SIG may include a PHY version identifier of 3 bits. For example, the PHY version identifier of 3 bits may include information related to a PHY version of a TX/RX PPDU. For example, a first value of the PHY version identifier of 3 bits may indicate that the TX/RX PPDU is an EHT PPDU. In other words, when the transmitting STA transmits the EHT PPDU, the PHY version identifier of 3 bits may be set to a first value. In other words, the receiving STA may determine that the RX PPDU is the EHT PPDU, based on the PHY version identifier having the first value.

(150) For example, the version-independent bits of the U-SIG may include a UL/DL flag field of 1 bit. A first value of the UL/DL flag field of 1 bit relates to UL communication, and a second value of the UL/DL flag field relates to DL communication.

(151) For example, the version-independent bits of the U-SIG may include information related to a TXOP length and information related to a BSS color ID.

(152) For example, when the EHT PPDU is divided into various types (e.g., various types such as

an EHT PPDU related to an SU mode, an EHT PPDU related to a MU mode, an EHT PPDU related to a TB mode, an EHT PPDU related to extended range transmission, or the like), information related to the type of the EHT PPDU may be included in the version-dependent bits of the U-SIG.

(153) For example, the U-SIG may include: 1) a bandwidth field including information related to a bandwidth; 2) a field including information related to an MCS scheme applied to EHT-SIG; 3) an indication field including information regarding whether a dual subcarrier modulation (DCM) scheme is applied to EHT-SIG; 4) a field including information related to the number of symbol used for EHT-SIG; 5) a field including information regarding whether the EHT-SIG is generated across a full band; 6) a field including information related to a type of EHT-LTF/STF; and 7) information related to a field indicating an EHT-LTF length and a CP length.

(154) Preamble puncturing may be applied to the PPDU of FIG. 14. The preamble puncturing implies that puncturing is applied to part (e.g., a secondary 20 MHz band) of the full band. For example, when an 80 MHz PPDU is transmitted, an STA may apply puncturing to the secondary 20 MHz band out of the 80 MHz band, and may transmit a PPDU only through a primary 20 MHz band and a secondary 40 MHz band.

(155) For example, a pattern of the preamble puncturing may be configured in advance. For example, when a first puncturing pattern is applied, puncturing may be applied only to the secondary 20 MHz band within the 80 MHz band. For example, when a second puncturing pattern is applied, puncturing may be applied to only any one of two secondary 20 MHz bands included in the secondary 40 MHz band within the 80 MHz band. For example, when a third puncturing pattern is applied, puncturing may be applied to only the secondary 20 MHz band included in the primary 80 MHz band within the 160 MHz band (or 80+80 MHz band). For example, when a fourth puncturing is applied, puncturing may be applied to at least one 20 MHz channel not belonging to a primary 40 MHz band in the presence of the primary 40 MHz band included in the 80 MHz band within the 160 MHz band (or 80+80 MHz band).

(156) Information related to the preamble puncturing applied to the PPDU may be included in U-SIG and/or EHT-SIG. For example, a first field of the U-SIG may include information related to a contiguous bandwidth, and second field of the U-SIG may include information related to the preamble puncturing applied to the PPDU.

(157) For example, the U-SIG and the EHT-SIG may include the information related to the preamble puncturing, based on the following method. When a bandwidth of the PPDU exceeds 80 MHz, the U-SIG may be configured individually in unit of 80 MHz. For example, when the bandwidth of the PPDU is 160 MHz, the PPDU may include a first U-SIG for a first 80 MHz band and a second U-SIG for a second 80 MHz band. In this case, a first field of the first U-SIG may include information related to a 160 MHz bandwidth, and a second field of the first U-SIG may include information related to a preamble puncturing (i.e., information related to a preamble puncturing pattern) applied to the first 80 MHz band. In addition, a first field of the second U-SIG may include information related to a 160 MHz bandwidth, and a second field of the second U-SIG may include information related to a preamble puncturing (i.e., information related to a preamble puncturing pattern) applied to the second 80 MHz band. Meanwhile, an EHT-SIG contiguous to the first U-SIG may include information related to a preamble puncturing applied to the second 80 MHz band (i.e., information related to a preamble puncturing pattern), and an EHT-SIG contiguous to the second U-SIG may include information related to a preamble puncturing (i.e., information related to a preamble puncturing pattern) applied to the first 80 MHz band.

(158) Additionally or alternatively, the U-SIG and the EHT-SIG may include the information related to the preamble puncturing, based on the following method. The U-SIG may include information related to a preamble puncturing (i.e., information related to a preamble puncturing pattern) for all bands. That is, the EHT-SIG may not include the information related to the preamble puncturing, and only the U-SIG may include the information related to the preamble puncturing

(i.e., the information related to the preamble puncturing pattern).

(159) The U-SIG may be configured in unit of 20 MHz. For example, when an 80 MHz PPDU is configured, the U-SIG may be duplicated. That is, four identical U-SIGs may be included in the 80 MHz PPDU. PPDUs exceeding an 80 MHz bandwidth may include different U-SIGs.

(160) The EHT-SIG of FIG. 13 may include control information for the receiving STA. The EHT-SIG may be transmitted through at least one symbol, and one symbol may have a length of 4 us. Information related to the number of symbols used for the EHT-SIG may be included in the U-SIG.

(161) The EHT-SIG may include a technical feature of the HE-SIG-B described with reference to FIG. 7 and FIG. 8. For example, the EHT-SIG may include a common field and a user-specific field as in the example of FIG. 7. The common field of the EHT-SIG may be omitted, and the number of user-specific fields may be determined based on the number of users.

(162) As in the example of FIG. 7, the common field of the EHT-SIG and the user-specific field of the EHT-SIG may be individually coded. One user block field included in the user-specific field may include information for two users, but a last user block field included in the user-specific field may include information for one user. That is, one user block field of the EHT-SIG may include up to two user fields. As in the example of FIG. 8, each user field may be related to MU-MIMO allocation, or may be related to non-MU-MIMO allocation.

(163) As in the example of FIG. 7, the common field of the EHT-SIG may include a CRC bit and a tail bit. A length of the CRC bit may be determined as 4 bits. A length of the tail bit may be determined as 6 bits, and may be set to '000000'.

(164) As in the example of FIG. 7, the common field of the EHT-SIG may include RU allocation information. The RU allocation information may imply information related to a location of an RU to which a plurality of users (i.e., a plurality of receiving STAs) are allocated. The RU allocation information may be configured in unit of 8 bits (or N bits), as in Table 1.

(165) The example of Table 5 to Table 7 is an example of 8-bit (or N-bit) information for various RU allocations. An index shown in each table may be modified, and some entries in Table 5 to Table 7 may be omitted, and entries (not shown) may be added.

(166) TABLE-US-00005 TABLE 5 Number Indices #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 0 26 26 26 26 26 26 26 1 1 26 26 26 26 26 26 26 52 1 2 26 26 26 26 26 26 52 26 26 1 3 26 26 26 26 26 52 52 1 4 26 26 52 26 26 26 26 26 1 5 26 26 52 26 26 26 52 1 6 26 26 52 26 52 26 26 1 7 26 26 52 26 52 52 1 8 52 26 26 26 26 26 26 26 1 9 52 26 26 26 26 26 52 1 10 52 26 26 26 52 26 26 1 11 52 26 26 26 52 52 1 12 52 52 26 26 26 26 26 26 1 13 52 52 26 26 26 52 1 14 52 52 26 52 26 26 1 15 52 52 26 52 52 1 16 26 26 26 26 26 106 1 17 26 26 52 26 106 1 18 52 26 26 26 106 1 19 52 52 26 106 1

(167) TABLE-US-00006 TABLE 6 Number Indices #1 #2 #3 #4 #5 #6 #7 #8 #9 of entries 20 106 26 26 26 26 26 1 21 106 26 26 26 52 1 22 106 26 52 26 26 1 23 106 26 52 52 1 24 52 52 — 52 52 1 25 242-tone RU empty (with zero users) 1 26 106 26 106 1 27-34 242 8 35-42 484 8 43-50 996 8 51-58 2*996 8 59 26 26 26 26 26 52 + 26 26 1 60 26 26 + 52 26 26 26 26 26 1 61 26 26 + 52 26 26 26 52 1 62 26 26 + 52 26 52 26 26 1 63 26 26 52 26 52 + 26 26 1 64 26 26 + 52 26 52 + 26 26 1 65 26 26 + 52 26 52 52 1

(168) TABLE-US-00007 TABLE 7 66 52 26 26 26 52 + 26 26 1 67 52 52 26 52 + 26 26 1 68 52 52 + 26 52 52 1 69 26 26 26 26 26 + 106 1 70 26 26 + 52 26 106 1 71 26 26 52 26 + 106 1 72 26 26 + 52 26 + 106 1 73 52 26 26 26 + 106 1 74 52 52 26 + 106 1 75 106 + 26 26 26 26 26 1 76 106 + 26 26 26 52 1 77 106 + 26 52 26 26 1 78 106 26 52 + 26 26 1 79 106 + 26 52 + 26 26 1 80 106 + 26 52 52 1 81 106 + 26 106 1 82 106 26 + 106 1

(169) The example of Table 5 to Table 7 relates to information related to a location of an RU allocated to a 20 MHz band. For example, 'an index 0' of Table 5 may be used in a situation where nine 26-RUs are individually allocated (e.g., in a situation where nine 26-RUs shown in FIG. 5 are individually allocated).

(170) Meanwhile, a plurality of RUs may be allocated to one STA in the EHT system. For example,

regarding 'an index 60' of Table 6, one 26-RU may be allocated for one user (i.e., receiving STA) to the leftmost side of the 20 MHz band, one 26-RU and one 52-RU may be allocated to the right side thereof, and five 26-RUs may be individually allocated to the right side thereof.

(171) A mode in which the common field of the EHT-SIG is omitted may be supported. The mode in which the common field of the EHT-SIG is omitted may be called a compressed mode. When the compressed mode is used, a plurality of users (i.e., a plurality of receiving STAs) may decode the PPDU (e.g., the data field of the PPDU), based on non-OFDMA. That is, the plurality of users of the EHT PPDU may decode the PPDU (e.g., the data field of the PPDU) received through the same frequency band. Meanwhile, when a non-compressed mode is used, the plurality of users of the EHT PPDU may decode the PPDU (e.g., the data field of the PPDU), based on OFDMA. That is, the plurality of users of the EHT PPDU may receive the PPDU (e.g., the data field of the PPDU) through different frequency bands.

(172) The EHT-SIG may be configured based on various MCS schemes. As described above, information related to an MCS scheme applied to the EHT-SIG may be included in U-SIG. The EHT-SIG may be configured based on a DCM scheme. For example, among N data tones (e.g., 52 data tones) allocated for the EHT-SIG, a first modulation scheme may be applied to half of consecutive tones, and a second modulation scheme may be applied to the remaining half of the consecutive tones. That is, a transmitting STA may use the first modulation scheme to modulate specific control information through a first symbol and allocate it to half of the consecutive tones, and may use the second modulation scheme to modulate the same control information by using a second symbol and allocate it to the remaining half of the consecutive tones. As described above, information (e.g., a 1-bit field) regarding whether the DCM scheme is applied to the EHT-SIG may be included in the U-SIG.

(173) An HE-STF of FIG. 13 may be used for improving automatic gain control estimation in a multiple input multiple output (MIMO) environment or an OFDMA environment. An HE-LTF of FIG. 13 may be used for estimating a channel in the MIMO environment or the OFDMA environment.

(174) A PPDU (e.g., EHT-PPDU) of FIG. 13 may be configured based on the example of FIG. 4 and FIG. 5.

(175) For example, an EHT PPDU transmitted on a 20 MHz band, i.e., a 20 MHz EHT PPDU, may be configured based on the RU of FIG. 4. That is, a location of an RU of EHT-STF, EHT-LTF, and data fields included in the EHT PPDU may be determined as shown in FIG. 4.

(176) An EHT PPDU transmitted on a 40 MHz band, i.e., a 40 MHz EHT PPDU, may be configured based on the RU of FIG. 5. That is, a location of an RU of EHT-STF, EHT-LTF, and data fields included in the EHT PPDU may be determined as shown in FIG. 5.

(177) Since the RU location of FIG. 5 corresponds to 40 MHz, a tone-plan for 80 MHz may be determined when the pattern of FIG. 6 is repeated twice. That is, an 80 MHz EHT PPDU may be transmitted based on a new tone-plan in which not the RU of FIG. 6 but the RU of FIG. 5 is repeated twice.

(178) When the pattern of FIG. 5 is repeated twice, 23 tones (i.e., 11 guard tones+12 guard tones) may be configured in a DC region. That is, a tone-plan for an 80 MHz EHT PPDU allocated based on OFDMA may have 23 DC tones. Unlike this, an 80 MHz EHT PPDU allocated based on non-OFDMA (i.e., a non-OFDMA full bandwidth 80 MHz PPDU) may be configured based on a 996-RU, and may include 5 DC tones, 12 left guard tones, and 11 right guard tones.

(179) A tone-plan for 160/240/320 MHz may be configured in such a manner that the pattern of FIG. 5 is repeated several times.

(180) The PPDU of FIG. 13 may be determined (or identified) as an EHT PPDU based on the following method.

(181) A receiving STA may determine a type of an RX PPDU as the EHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the EHT PPDU: 1) when a

first symbol after an L-LTF signal of the RX PPDU is a BPSK symbol; 2) when RL-SIG in which the L-SIG of the RX PPDU is repeated is detected; and 3) when a result of applying “module 3” to a value of a length field of the L-SIG of the RX PPDU is detected as “0”. When the RX PPDU is determined as the EHT PPDU, the receiving STA may detect a type of the EHT PPDU (e.g., an SU/MU/Trigger-based/Extended Range type), based on bit information included in a symbol after the RL-SIG of FIG. 13. In other words, the receiving STA may determine the RX PPDU as the EHT PPDU, based on: 1) a first symbol after an L-LTF signal, which is a BPSK symbol; 2) RL-SIG contiguous to the L-SIG field and identical to L-SIG; 3) L-SIG including a length field in which a result of applying “modulo 3” is set to “0”; and 4) a 3-bit PHY version identifier of the aforementioned U-SIG (e.g., a PHY version identifier having a first value).

(182) For example, the receiving STA may determine the type of the RX PPDU as the EHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the HE PPDU: 1) when a first symbol after an L-LTF signal is a BPSK symbol; 2) when RL-SIG in which the L-SIG is repeated is detected; and 3) when a result of applying “module 3” to a value of a length field of the L-SIG is detected as “1” or “2.”

(183) For example, the receiving STA may determine the type of the RX PPDU as a non-HT, HT, and VHT PPDU, based on the following aspect. For example, the RX PPDU may be determined as the non-HT, HT, and VHT PPDU: 1) when a first symbol after an L-LTF signal is a BPSK symbol; and 2) when RL-SIG in which L-SIG is repeated is not detected. In addition, even if the receiving STA detects that the RL-SIG is repeated, when a result of applying “modulo 3” to the length value of the L-SIG is detected as “0,” the RX PPDU may be determined as the non-HT, HT, and VHT PPDU.

(184) In the following example, a signal represented as a (TX/RX/UL/DL) signal, a (TX/RX/UL/DL) frame, a (TX/RX/UL/DL) packet, a (TX/RX/UL/DL) data unit, (TX/RX/UL/DL) data, or the like may be a signal transmitted/received based on the PPDU of FIG. 13. The PPDU of FIG. 13 may be used to transmit/receive frames of various types. For example, the PPDU of FIG. 13 may be used for a control frame. An example of the control frame may include a request to send (RTS), a clear to send (CTS), a power save-poll (PS-poll), BlockACKReq, BlockAck, a null data packet (NDP) announcement, and a trigger frame. For example, the PPDU of FIG. 14 may be used for a management frame. An example of the management frame may include a beacon frame, a (re-)association request frame, a (re-)association response frame, a probe request frame, and a probe response frame. For example, the PPDU of FIG. 13 may be used for a data frame. For example, the PPDU of FIG. 13 may be used to simultaneously transmit at least two or more of the control frame, the management frame, and the data frame.

(185) FIG. 14 illustrates an example of a modified transmission apparatus and/or receiving apparatus of the present specification.

(186) Each apparatus/STA of the sub-figure (a)/(b) of FIG. 1 may be modified as shown in FIG. 15. A transceiver 630 of FIG. 14 may be identical to the transceivers 113 and 123 of FIG. 1. The transceiver 630 of FIG. 14 may include a receiver and a transmitter.

(187) A processor 610 of FIG. 14 may be identical to the processors 111 and 121 of FIG. 1. Alternatively, the processor 610 of FIG. 14 may be identical to the processing chips 114 and 124 of FIG. 1.

(188) A memory 620 of FIG. 14 may be identical to the memories 112 and 122 of FIG. 1. Alternatively, the memory 620 of FIG. 14 may be a separate external memory different from the memories 112 and 122 of FIG. 1.

(189) Referring to FIG. 14, a power management module 611 manages power for the processor 610 and/or the transceiver 630. A battery 612 supplies power to the power management module 611. A display 613 outputs a result processed by the processor 610. A keypad 614 receives inputs to be used by the processor 610. The keypad 614 may be displayed on the display 613. A SIM card 615 may be an integrated circuit which is used to securely store an international mobile subscriber

identity (IMSI) and its related key, which are used to identify and authenticate subscribers on mobile telephony devices such as mobile phones and computers.

(190) Referring to FIG. 14, a speaker 640 may output a result related to a sound processed by the processor 610. A microphone 641 may receive an input related to a sound to be used by the processor 610.

(191) Hereinafter, technical features applicable to the EHT standard will be described.

(192) According to an embodiment of the present specification, the EHT standard may support PPDU of 320 MHz bandwidth and 160+160 MHz. In addition, 240 MHz transmission and 160+80 MHz transmission may be supported. The 240 MHz transmission and 160+80 MHz transmission may be configured by applying 80 MHz preamble puncturing in 320 MHz bandwidth and 160+160 MHz bandwidth, respectively. For example, the 240 MHz bandwidth and 160+80 MHz bandwidth may be configured based on three 80 MHz channels including a primary 80 MHz (channel).

(193) According to an embodiment of the present specification, the EHT standard may reuse a tone plan of the IEEE 802.11ax standard a 20/40/80/160/80+80 MHz PPDU. According to an embodiment, a 160 MHz OFDMA tone plan of the IEEE 802.11ax standard may be duplicated and used for 320 MHz and 160+160 MHz PPDU.

(194) According to an embodiment of the present specification, the transmission in 240 MHz and 160+80 MHz may consist of three 80 MHz segments. For example, the tone plan of each 80 MHz segment may be configured in the same manner as the 80 MHz tone plan of the IEEE 802.11ax standard.

(195) According to an embodiment of the present specification, a 160 MHz tone plan may be duplicated and used for a non-OFDMA tone plan of a 320/160+160 MHz PPDU.

(196) According to an embodiment of the present specification, a duplicated HE160 tone plan may be used for a 320/160+160 MHz PPDU non-OFDMA tone plan.

(197) According to an embodiment of the present specification, in each 160 MHz segment for a non-OFDMA tone plan of a 320/160+160 MHz PPDU, 12 and 11 null tones may be configured on the leftmost side and the rightmost side, respectively.

(198) According to an embodiment of the present specification, the data part of the EHT PPDU may use the same subcarrier spacing as the data part of the IEEE 802.11ax standard.

(199) Hereinafter, technical features of a resource unit (RU) applicable to the EHT standard will be described.

(200) According to an embodiment of the present specification, in the EHT standard, one or more RUs may be allocated to a single STA. For example, coding and interleaving schemes for multiple RUs allocated to a single STA may be variously set.

(201) According to an embodiment of the present specification, small-size RUs may be aggregated with other small-size RUs. According to an embodiment of the present specification, large-size RUs may be aggregated with other large-size RUs.

(202) For example, RUs of 242 tones or more may be defined/set as 'large size RUs'. For another example, RUs of less than 242 tones may be defined/configured as 'small size RUs'.

(203) According to an embodiment of the present specification, there may be one PSDU per STA for each link. According to an embodiment of the present specification, for LDPC encoding, one encoder may be used for each PSDU.

(204) Small-Size RUs

(205) According to an embodiment of the present specification, an aggregation of small-size RUs may be set so as not to cross a 20 MHz channel boundary. For example, RU106+RU26 and RU52+RU26 may be configured as an aggregation of small-size RUs.

(206) According to an embodiment of the present specification, in PPDU of 20 MHz and 40 MHz, contiguous RU26 and RU106 may be aggregated/combined within a 20 MHz boundary.

(207) According to an embodiment of the present specification, in PPDU of 20 MHz and 40 MHz, RU26 and RU52 may be aggregated/combined.

(208) For example, in 20 MHz (or 20 MHz PPDU), an example of contiguous RU26 and RU52 may be shown through FIG. 21.

(209) FIG. 15 shows an example of an aggregation of RU26 and RU52 in 20 MHz.

(210) Referring to FIG. 15, shaded RU26 and RU52 may be aggregated. For example, the second RU26 and the second RU52 may be aggregated. For another example, the seventh RU and the third RU52 may be aggregated.

(211) For example, in 40 MHz, an example of contiguous RU26 and RU52 is described in FIG. 15.

(212) FIG. 16 shows an example of an aggregation of RU26 and RU52 in 40 MHz.

(213) Referring to FIG. 16, shaded RU26 and RU52 may be aggregated. For example, the second RU26 and the second RU52 may be aggregated. For another example, the eighth RU26 and the third RU52 may be aggregated. For another example, the eleventh RU26 and the sixth RU52 may be aggregated. For another example, the seventeenth RU26 and the seventh RU52 may be aggregated.

(214) According to an embodiment of the present specification, RU26 and RU52 may be aggregated/combined in a PPDU of 80 MHz.

(215) For example, an example of contiguous RU26 and RU52 in 80 MHz may be shown by FIG. 17.

(216) FIG. 17 shows an example of an aggregation of RU26 and RU52 in 80 MHz.

(217) Referring to FIG. 17, 80 MHz may be divided into the first 40 MHz and the second 40 MHz. For example, within the first 40 MHz, the 8th RU26 and the 3rd RU52 may be aggregated. For another example, within the first 40 MHz, the 11th RU26 and the 6th RU52 may be aggregated. For another example, within the second 40 MHz, the 8th RU26 and the 3rd RU52 may be aggregated. For another example, within the second 40 MHz, the 11th RU26 and the 6th RU52 may be aggregated.

(218) According to an embodiment, when LDPC coding is applied, a single tone mapper may be used for RUs having less than 242 tones.

(219) Large-Size RUs

(220) According to an embodiment, in OFDMA transmission of 320/160+160 MHz for a single STA, an aggregation of a large-size RUs may be allowed only within a primary 160 MHz or a secondary 160 MHz. For example, the primary 160 MHz (channel) may consist of a primary 80 MHz (channel) and a secondary 80 MHz (channel). The secondary 160 MHz (channel) can be configured with channels other than the primary 160 MHz.

(221) According to an embodiment, in OFDMA transmission of 240 MHz for a single STA, an aggregated of large-size RUs may be allowed only within 160 MHz (band/channel), and the 160 MHz may consist of two adjacent 80 MHz channels.

(222) According to an embodiment, in OFDMA transmission of 160+80 MHz for a single STA, an aggregation of large-size RUs may be allowed only within a continuous 160 MHz (band/channel) or within the remaining 80 MHz (band/channel).

(223) In 160 MHz OFDMA, an aggregation of large-size RUs configured as shown in Table 8 may be supported.

(224) TABLE-US-00008 TABLE 8 RU size Aggregate BW Notes 484 + 996 120 MHz 4 options

(225) In 80 MHz OFDMA, an aggregation of large-size RUs configured as shown in Table 9 may be supported.

(226) TABLE-US-00009 TABLE 9 RU size Aggregate BW Notes 484 + 242 60 MHz 4 options

(227) In 80 MHz non-OFDMA, an aggregation of large-size RUs configured as shown in Table 10 may be supported. In 80 MHz non-OFDMA, puncturing can be applied. For example, one of four 242 RUs may be punctured.

(228) TABLE-US-00010 TABLE 10 RU size Aggregate BW Notes 484 + 242 60 MHz 4 options

(229) In 160 MHz non-OFDMA, an aggregation of large-size RUs configured as shown in Table 11 may be supported. In 160 MHz non-OFDMA, puncturing can be applied. For example, one of eight

242 RUs may be punctured. For another example, one of four 484 RUs may be punctured.

(230) TABLE-US-00011 TABLE 11 80 MHz RU Size 80 MHz RU size Aggregate BW Notes 484 996 120 MHz 4 options 484 + 242 996 140 MHz 8 options

(231) In 240 MHz non-OFDMA, an aggregation of large-size RUs configured as shown in Table 12 may be supported. In 240 MHz non-OFDMA, puncturing can be applied. For example, one of six 484 RUs may be punctured. For another example, one of three 996 RUs may be punctured.

(232) TABLE-US-00012 TABLE 12 80 MHz 80 MHz 80 MHz RU size RU size RU size Aggregate BW Notes 484 996 996 200 MHz 6 options — 996 996 160 MHz 3 options

(233) In 320 MHz non-OFDMA, an aggregation of large-size RUs configured as shown in Table 13 may be supported. In 320 MHz non-OFDMA, puncturing can be applied. For example, one of eight 484 RUs may be punctured. For another example, one of four 996 RUs may be punctured.

(234) TABLE-US-00013 TABLE 13 80 MHz 80 MHz 80 MHz 80 MHz RU size RU size RU size RU size Aggregate BW Notes 484 996 996 996 280 MHz 8 options — 996 996 996 240 MHz 4 options

(235) Hereinafter, technical features related to the operating mode will be described.

(236) According to an embodiment, a station (STA) supporting the EHT standard STA (hereinafter, “EHT STA”) or a station (STA) supporting the EHT standard STA (hereinafter, “HE STA”) may operate in a 20 MHz channel width mode. In the 20 MHz channel width mode, the EHT STA may operate by reducing the operating channel width to 20 MHz using an operating mode indication (OMI).

(237) According to an embodiment, the EHT STA (or HE STA) may operate in an 80 MHz channel width mode. For example, in the 80 MHz channel width mode, the EHT STA may operate by reducing the operating channel width to 80 MHz using an operating mode indication (OMI).

(238) According to an embodiment, the EHT STA may support sub-channel selective transmission (SST). An STA supporting the SST can quickly select (and switch to) another channel between transmissions to cope with fading in a narrow sub-channel.

(239) The 802.11be standard (i.e., the EHT standard) can provide a higher data rate than the 802.11ax standard. The EHT (i.e., extreme high throughput) standard can support wide bandwidth (up to 320 MHz), 16 streams, and multi-band operation.

(240) In the EHT standard, various preamble puncturing or multiple RU allocation may be supported in wide bandwidth (up to 320 MHz) and SU/MU transmission. In addition, in the EHT standard, a signal transmission/reception method through 80 MHz segment allocation is considered in order to support an STA with low end capability (e.g., 80 MHz only operating STA).

Accordingly, in the following specification, a method of configuring/transmitting an EHT-SIG for the MU transmission in consideration of sub-channel selective transmission (SST) defined in the 11ax standard and Multi-RU aggregation may be proposed. For example, the EHT-SIG may be configured as a self-contained EHT-SIG. When the self-contained EHT-SIG is used, a technical feature for signaling RU allocation may be proposed in the present specification.

(241) EHT PPDU Configuration

(242) In order to support a transmission method based on the EHT standard, a new frame format may be used. When transmitting a signal through the 2.4/5/6 GHz band based on the new frame format, conventional Wi-Fi receivers (or STAs) (e.g., 802.11n) as well as receivers supporting the EHT standard receivers in compliance with the 802.11n/ac/ax standard) can also receive EHT signals transmitted through the 2.4/5/6 GHz band.

(243) The preamble of the PPDU based on the EHT standard can be set in various ways.

Hereinafter, an embodiment of configuring the preamble of the PPDU based on the EHT standard will be described. Hereinafter, a PPDU based on the EHT standard may be described as an EHT PPDU. However, the EHT PPDU is not limited to the EHT standard. The EHT PPDU may include not only the 802.11be standard (i.e., the EHT standard), but also a PPDU based on a new standard that is improved/evolved/extended with the 802.11be standard.

(244) FIG. 18 shows an example of an EHT PPDU.

(245) Referring to FIG. 18, an EHT PPDU 1800 may include an L-part 1810 and an EHT-part 1820. The EHT PPDU 1800 may be configured in a format to support backward compatibility. In addition, the EHT PPDU 1800 may be transmitted to a single STA and/or multiple STAs. The EHT PPDU 1800 may be an example of an MU-PPDU of the EHT standard.

(246) The EHT PPDU 1800 may include the L-part 1810 preceding the EHT-part 1820 for coexistence or backward compatibility with a legacy STA (e.g., STA in compliance with the 802.11n/ac/ax standard). For example, the L-part 1810 may include L-STF, L-LTF, and L-SIG. For example, phase rotation may be applied to the L-part 1810.

(247) According to an embodiment, the EHT part 1820 may include RL-SIG, U-SIG 1821, EHT-SIG 1822, EHT-STF, EHT-LTF, and data fields. Similar to the 11ax standard, RL-SIG may be included in the EHT part 1820 for L-SIG reliability and range extension. The RL-SIG may be transmitted immediately after the L-SIG, and may be configured to repeat the L-SIG.

(248) For example, four additional subcarriers may be applied to L-SIG and RL-SIG. The extra subcarriers may be configured at subcarrier indices $[-28, -27, 27, 28]$. The extra subcarriers may be modulated in a BPSK scheme. In addition, coefficients of $[-1 \ -1 \ -1 \ 1]$ may be mapped to the extra subcarriers.

(249) For example, the EHT-LTF may be one of $1 \times$ EHT-LTF, $2 \times$ EHT-LTF, or $4 \times$ EHT-LTF. The EHT standard may support EHT-LTF for 16 spatial streams.

(250) Each field in FIG. 18 may be the same as corresponding field described in FIG. 13.

(251) Hereinafter, technical features that can be further improved in the present specification will be described.

(252) The present specification may improve the technical features of various types of control information (or control fields). For example, the present specification may improve the technical characteristics of the EHT-SIG field (or EHT-SIG signal). More specifically, the present specification may improve the common field of the EHT-SIG field.

(253) For example, the WLAN system according to the present specification (e.g., the 11be standard) supports wide bandwidth communication supporting a maximum bandwidth of 320 MHz, and various types of preamble puncturing and/or multiple RU allocation may be supported. In addition, the 11be standard may support a signal transmission/reception method through 80 MHz segment allocation in order to support an STA having a low end capability (e.g., 80 MHz only operating STA). Considering these technical characteristics, when the 11be MU-PPDU for signal transmission and reception of multiple users is used, various control fields included in the EHT-SIG may include common control information not included in the U-SIG. For example, the present specification proposes various examples of including common control information, which is not included in the U-SIG, in a common field of the EHT field.

(254) In other words, an example of the present specification proposes various transmit/receive PPDUs. An example of the transmission/reception (TX/RX) PPDU may include various fields described in FIGS. 3, 7, 8, 13, 18, and 19. More specifically, an example of the transmission/reception PPDU may include at least one legacy field (e.g., L-STF, L-LTF, L-SIG, and RL-SIG in FIG. 18). In addition, an example of the transmission/reception PPDU may include a first control signal field (e.g., U-SIG field) and a second control signal field (e.g., EHT-SIG field) for the transmission/reception PPDU. For example, the first control signal field may be the U-SIG 1821 of FIG. 18, and the second control signal field may be the EHT-SIG 1822 of FIG. 18.

(255) Control information not included in the first control signal field (e.g., U-SIG field) may be called various names such as 'overflowed information' or 'overflow information'. The second control signal field (e.g., EHT-SIG field) may include a common field and a user specific field. Each of the common field and user specific field may include at least one encoding block (e.g., a binary convolutional code (BCC) encoding block). One encoding block may be transmitted/received through at least one symbol, and one encoding block is not necessarily

transmitted through one symbol. Meanwhile, one symbol for transmitting an encoding block may have a symbol length of 4 μ s.

(256) According to an example of the present specification, the overflowed information may be included in a common field of the second control signal field (e.g., EHT-SIG field). The present specification proposes various examples of the overflowed information. In addition, the present specification proposes various examples of the common field and/or user specific field. A specific example is as follows.

(257) The TX/RX PPDU proposed in the present specification may be used for communication for at least one user. For example, the technical features of the present specification may be applied to an MU-PPDU (e.g., EHT MU PPDU) conforming to the 11be standard. For example, an example of an MU-PPDU for transmitting a signal to multiple STAs in consideration of backward compatibility may be the PPDU of FIG. 18.

(258) FIG. 19 shows an example of a first control signal field or U-SIG field of the present specification.

(259) As depicted, the first control signal field (e.g., U-SIG field) may include a version independent field **1910** and a version dependent field **1920**. For example, the version independent field **1910** may include control information that is constantly included regardless of the version of the WLAN (e.g., the IEEE 802.11 be and next-generation standards of the IEEE 802.11be). For example, the version dependent field **1920** may include control information dependent on a corresponding version (e.g., the IEEE 802.11be).

(260) For example, the version independent field **1910** may include information related to a 3-bit version identifier indicating a Wi-Fi version defined after the 11be and 11be standards, a 1-bit DL/UL field BSS color, and/or a TXOP duration. For example, the version dependent field **1920** may include information related to a PPDU format type, a Bandwidth, and/or an MCS.

(261) For example, in the first control signal field (e.g., U-SIG field) shown in FIG. 19, two symbols (e.g., two contiguous symbols each having a 4 μ s length) may be jointly encoded. In addition, the field of FIG. 19 may be configured based on 52 data tones and 4 pilot tones for each 20 MHz band/channel. In addition, the field of FIG. 19 may be modulated in the same manner as the HE-SIG-A of the conventional 11ax standard. In other words, the field of FIG. 19 may be modulated based on a BPSK scheme with a 1/2 code rate.

(262) For example, the second control signal field (e.g., EHT-SIG field) may be divided into a common field and a user specific field, and may be encoded based on various MCS levels. For example, the common field may include indication information related to a spatial stream used in a transmission/reception PPDU (e.g., a data field) and indication information related to an RU. For example, the user specific field may include ID information used by at least one specific user (or receiving STA), MCS, and indication information related to coding. In other words, the user specific field includes decoding information (e.g., corresponding to the data field transmitted through at least one RU indicated by an RU allocation sub-field included in the common field). STA ID information assigned to the RU, MSC information, and/or channel coding type/rate information). In other words, the user specific field may include decoding information (STA ID information, MSC information, and/or channel coding type/rate information allocated to a corresponding RU) for a data field transmitted through at least one RU indicated by an RU allocation sub-field included in the common field.

(263) An example of an information field/bit that may be included in the first control signal field (e.g., U-SIG field) is shown in Table 14 below. As described below, since there is a restriction on the length of the first control signal field (e.g., U-SIG field), some of the fields in Table 14 may overflow into other fields. That is, the bit lengths described in the table below may be changed, and at least one of the individual fields/bits listed in the table below may be omitted. Also, other fields/bits may be added.

(264) TABLE-US-00014 TABLE 14 Field bits PHY version Identifier 3 TXOP 7 BSS Color 6

DL/UL 1 BW 3 PPDU format 2 EHT-SIG MCS 3 Nsym of EHT-SIG/users of MU-MIMO 5 GI + LTF 2 Coding 1 LDPC Extra symbol 1 STBC 1 Beamformed 1 Pre-FEC padding 2 PE Disambiguity 1 doppler 1 spatial reuse 4 beam change 1 DCM 1 HARQ 1 Multi-AP 1 Compression 1 CRC 4 Tail 6 Total bits 54

(265) The first control signal field (e.g., U-SIG field) may consist of two contiguous symbols. In this case, the maximum number of bits that can be included in the first control signal field (e.g., U-SIG field) may be fixed or preset (e.g., fixed to 48/52 bits or preset). Accordingly, information that is not included in the first control signal field (e.g., U-SIG field) may exist. Such information/field may be referred to as various names, such as overflowed information, overflow information, U-SIG overflow, and U-SIG overflow information/field. According to an example of the present specification, the overflowed information is preferably included in the second control signal field (e.g., EHT-SIG field). In addition, since the overflowed information may not be user specific information, the corresponding information may be preferably included in the common field of the second control signal field (e.g., EHT-SIG field).

(266) For example, an example of a control field included in the first control signal field (e.g., U-SIG field) of the present specification may be based on Tables 15 and 16. Fields/bits not shown in Tables 15 and 16 may be the overflowed information.

(267) However, the technical features of the present specification are not limited to the following examples. Accordingly, the control field and/or overflowed information included in the first control signal field (e.g., U-SIG field) may be configured in various ways.

(268) TABLE-US-00015 TABLE 15 Index contents bits 1 PHY version identifier 3 2 UL/DL flag 1 3 TXOP 7 4 BSS Color 6 5 BW 4 6 PPDU type 2 7 Compression mode 1 8 GI + EHT-LTF Size 2

(269) TABLE-US-00016 TABLE 16 Index contents bits 9 Spatial reuse 4 10 EHT-SIG MCS (including DCM) 3 11 Number of EHT-SIG Symbols/ 5 Number of users for MU-MIMO 12 LDPC ExtraSymbol Segment 1 13 Pre-FECPadding Factor 2 14 PE Disambiguity 1 15 CRC 4 16 Tail 6

(270) As described above, the overflowed information of the present specification is preferably included in a common field of the second control signal field (e.g., EHT-SIG field). Hereinafter, technical characteristics of the common field of the second control signal field (e.g., EHT-SIG field) will be described.

(271) For example, the common field of the second control signal field (e.g., EHT-SIG field) is an RU allocation sub-field, a CRC field, and/or a Tail field in addition to the overflowed information. 'N' RU allocation sub-fields may be included, and one RU allocation sub-field may have a length of 9 bits (or 10 bits). One RU allocation subfield may include allocation information (or location information) of at least one RU included in an RU region corresponding to 20 MHz. For example, the one RU allocation subfield may be used for the same purpose/purpose as the 8-bit allocation information of the conventional 11ax standard which is described in the above Tables 1 and 2.

(272) For example, the second control signal field (e.g., EHT-SIG field) may be transmitted through a plurality of content channels (e.g., EHT-SIG content channel). In this case, it is preferable that the overflowed information included in the second control signal field (e.g., EHT-SIG field) is identically repeated to the plurality of content channels (e.g., EHT-SIG content channel). In other words, when the content channel (e.g., EHT-SIG content channel) is set to two (2), it is preferable that the overflowed information is identically included/repeated in both content channel 1 and content channel 2.

(273) FIG. 20 shows an example in which the overflowed information of this specification is repeated in two content channels.

(274) An example of FIG. 20 is an example of the second control signal field (e.g., EHT-SIG field) applied to a PPDU having a bandwidth of 80 MHz. As shown, the EHT-SIG field of FIG. 20 includes a Common field (2010) and a User Specific field (2020). In addition, the overflowed information described above is denoted as "U-SIG Overflow" in FIG. 20. As shown in FIG. 20, a PPDU having a bandwidth of 80 MHz is transmitted/received based on two content channels (i.e.,

EHT-SIG content channel 1 (2001) and EHT-SIG content channel 2 (2002)). Also, each content channel has a bandwidth of 20 MHz, and each of the two content channels can be duplicated in the frequency domain for the 80 MHz bandwidth. In this case, the overflowed information shown in FIG. 20, that is, “U-SIG Overflow” is included in EHT-SIG content channel 1 (2001) and equally/identically included in EHT-SIG content channel 2 (2002). That is, the common field included in the EHT-SIG content channel 1 (2001) and the common field included in the EHT-SIG content channel 2 (2002) do not include the same information, but overflow information, that is, “U-SIG Overflow” can be included in common. In other words, the EHT-SIG content channel 1 (2001) includes allocation information for the first RU region (RUs overlapping with tone ranges -500:-259 and 12:253), and the EHT-SIG content channel 2 (2002) includes allocation information for the second RU region (e.g., RUs overlapping with tone ranges -253:-12 and 259:500), but it is preferable that both of the two content channels (2001, 2002) include overflowed information, which is denoted as “U-SIG Overflow,” in common.

(275) The example of FIG. 20 relates to an 80 MHz PPDU, but the technical features of this specification can be applied to PPDU of various bandwidths. For example, it is preferable that the “U-SIG Overflow” is commonly included in both EHT-SIG content channel 1 and EHT-SIG content channel 2 for the 40 MHz PPDU. In addition, it is preferable that the “U-SIG Overflow” is commonly included in both EHT-SIG content channel 1 and EHT-SIG content channel 2 for 160 MHz or 320 MHz PPDU. In other words, in order to transmit/receive the second control signal field (e.g., EHT-SIG field) included in the 40/80/160/320 MHz PPDU of this specification, it is preferably that two content channels are defined (i.e., EHT-SIG content channel 1 (2001) and EHT-SIG content channel 2 (2002)). Each content channel may have a bandwidth of 20 MHz and may be duplicated in frequency. In this case, Common fields transmitted through the two content channels preferably includes “U-SIG Overflow” in common, and when the two content channels are duplicated, it is preferable that the same “U-SIG Overflow” is included in all common fields of the duplicated content channels. In addition, one of the two content channels includes RU allocation information for the first RU region (e.g., RUs overlapping with tone ranges -500:-259 and 12:253), and the other one preferably includes RU allocation information for the second RU region (e.g., RUs overlapping with tone ranges -253:-12 and 259:500).

(276) Technical features that can be additionally applied to the above-described technical features are described below. The individual technical features below can be combined in various ways.

(277) Technical Feature 1: Hereinafter, technical characteristics applicable to the common field of a content channel (e.g., EHT-SIG content channel) will be described.

(278) Technical Feature 1.A: FIG. 21 shows a common field of a control signal field according to an example of the present specification. The illustrated common field may mean a common field included in a content channel (e.g., EHT-SIG content channel) through which the second control signal field (e.g., EHT-SIG field) is transmitted.

(279) For example, the common fields of FIG. 21 may include the Overflowed information from the U-SIG, at least one RU allocation subfield, a CRC field, and a tail field. The RU allocation subfield may have a length of 9 bits and may be generated as many as N, the CRC field may have a length of 4 bits, and the Tail field may have a length of 6 bits.

(280) Encoding may be applied to the common field of FIG. 21 based on a binary convolutional code (BCC) scheme. For example, the common field may consist of one BCC block. In this specification, one BCC block may be included in at least one symbol (e.g., one symbol having a length of 4 μ s). For example, one BCC block may be transmitted and received through two contiguous symbols (e.g., one symbol having a length of 4 μ s).

(281) The length of the CRC field may be set longer than 4 bits. For example, the length of the CRC field may be one of 5, 6, 7, and 8 bits. Since U-SIG overflow is additionally included in the common field compared to the conventional standard, a long CRC field may be required to improve error detection capability through CRC.

(282) Technical Feature 2: The above example can be modified in various ways. For example, information overflowed from the U-SIG may be included in an independent block field in the EHT-SIG. In this case, the independent block field may include a tail field (e.g., 6-bit information) and a CRC field (e.g., 4-bit information), and may be a separate BCC block.

(283) FIG. 22 shows an additional example of a common field of a control signal field according to an example of the present specification. The illustrated common field may mean a common field included in a content channel (e.g., EHT-SIG content channel) through which the second control signal field (e.g., EHT-SIG field) is transmitted.

(284) As shown, an independent BCC block (the first BCC block shown) may be configured with only overflowed information, a CRC field, and a Tail field. The common bits shown in FIG. 22 may include at least one Resource Unit (RU) allocation sub-field. As described above, one RU allocation subfield may have a length of 9/10 bits.

(285) The above example may be modified in various ways. For example, a BCC block including the overflowed information may not include a CRC field, but instead include a 1-bit parity bit.

(286) The example above may be further modified. For example, the above-described common bits may be included in one BCC block in a unit of 2/4 RA fields. In other words, one BCC block includes common bits, common bits include 2 or 4 RA fields, and one RA field includes the aforementioned 9-bit RU allocation subfield. In this case, one BCC block may include a CRC field (e.g., 4 bits) and a Tail field (e.g., 6 bits).

(287) In the case of configuring the BCC block as in the above-described example, the receiving STA has the advantage of immediately knowing the mode information included in the corresponding block by decoding the corresponding BCC block. However, a disadvantage in that overhead increases due to the CRC field and the Tail field included in each BCC block may occur.

(288) The IEEE 802.11be standard supports large BW (e.g., 240 MHz, 320 MHz). In this case, if RU allocation information is configured based on a 20 MHz content channel (e.g., EHT-SIG content channel) as in the existing 11ax standard, the number of resource unit allocation sub-fields included in one content channel (e.g., EHT-SIG content channel) may be up to 8. If the length of one RU allocation sub-field is 9 bits, a maximum of 72 (=9*8) bits may be required for RU allocation. When a 4-bit CRC field such as the conventional 11ax is used for such long bit information (for example, 72-bit information), the error correction performance for the received bit may be weakened and the error probability for the received bit may increase.

(289) In view of the foregoing, the present specification proposes an additional example as follows.

(290) Technical Feature 3: For FA and clear error detection, the RU allocation subfield of the common field may consist of, for example, 2/4 subfields. In this case, the common field may include at least one encoding block (e.g., BCC block), and each encoding block may include a CRC field and a tail field.

(291) Technical Feature 3.A: For example, two RU allocation sub-fields may be included in one encoding block.

(292) FIG. 23 shows an additional example of a common field of a control signal field according to an example of the present specification. The illustrated common field may mean a common field included in a content channel (e.g., EHT-SIG content channel) through which the second control signal field (e.g., EHT-SIG field) is transmitted.

(293) As shown, the first block of two contiguous BCC blocks may include Overflowed information, two RU allocation subfields (RA fields), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits). In addition, the second block of two contiguous BCC blocks does not include the overflowed information, and includes two RU allocation subfields (RA fields), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(294) Technical Feature 3.B: For example, an example in which four RA subfields are included in one block is also possible.

(295) FIG. 24 shows an additional example of a common field of a control signal field according to

an example of the present specification. The illustrated common field may mean a common field included in a content channel (e.g., EHT-SIG content channel) through which the second control signal field (e.g., EHT-SIG field) is transmitted.

(296) As shown, the first block of two contiguous BCC blocks may include the Overflowed information, 4 RU allocation subfields (RA fields), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits). In addition, the second block of two consecutive BCC blocks does not include the overflowed information, and includes four RU allocation subfields (RA fields), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(297) Technical Feature 3.B.i: Modifying the above example, the first block of two consecutive BCC blocks may include overflowed information, two RU allocation subfields (i.e., RA fields), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits). In addition, the second block of two contiguous BCC blocks does not include the Overflowed information, and includes four RU allocation subfields (i.e., RA field), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits). And the third block being contiguous to the second block includes two RU allocation subfields (RA fields), two user fields (e.g., each field including user-specific information), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(298) Technical Feature 3.B.ii: In the above example, when the number of RU allocation subfields (i.e., RA fields) is 4, the second encoding block of the common field may include 2 RU allocation subfields, 2 user fields (for example, each field including user-specific information), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(299) Technical Feature 3.C: The above example can be changed in various ways. For example, the second BCC block described in 'Technical Feature 3.A' and 'Technical Feature 3.B' described above includes the remaining RU allocation subfields (i.e., RA field) other than the RA field of the first BCC block. For example, the first BCC block shown in 'Technical Feature 3.A' includes two RU allocation subfields (i.e., RA fields), and the second BCC block shown in 'Technical Feature 3.A' includes the remaining RA field.

(300) Technical Feature 3.C.i: In the above example, the number of RU allocation subfields (i.e., RA fields) included in the first BCC block may be 1/2/3/4.

(301) Technical Feature 3.C.ii: In the above example, the number of BCC blocks configuring the EHT-SIG common field may be changed according to the number of BW and/or RU allocation subfields (i.e., RA fields) of the PPDU.

(302) Technical Feature 3.D: The number of RU allocation subfields (i.e., RA fields) described above may be variously changed as follows.

(303) Technical Feature 3.D.i: For example, when the number of RU allocation subfields (i.e., RA fields) included in one content channel (e.g., EHT-SIG content channel) is less than 4 (or less than or equal to 4), common field (i.e., the EHT-SIG common field) may be configured as in the above 'Technical Feature 3.A'. In other words, if the number of RU allocation subfields (i.e., RA fields) is less than 4 (or less than or equal to 4), the first BCC block may include overflowed information, two RU allocation subfields (i.e., RA fields), and a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(304) Technical Feature 3.D.i.1: Additionally or alternatively, when the number of RU allocation subfields (i.e., RA fields) included in one content channel (e.g., EHT-SIG content channel) is less than 4 (or less than or equal to 4), one content channel may consist of one BCC block. That is, a common channel included in one content channel (e.g., EHT-SIG content channel) is composed of one BCC block, and the corresponding one BCC block may include overflowed information, N RAs (i.e., total RU allocation subfields required for RU allocation), a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(305) Technical Feature 3.D.ii: Additionally or alternatively, when the number of RU allocation subfields (i.e., RA fields) included in one content channel (e.g., EHT-SIG content channel) is more than 4 (or more than or equal to 4), the common field (e.g., EHT-SIG common field) may consist of

two BCC blocks.

(306) Technical Feature 3.D.ii.1: In this case, the first block among the two BCC blocks may include overflowed information, two RU allocation subfields (i.e., RA fields), a CRC field (e.g., 4 bit), and a Tail field (e.g., 6 bits).

(307) Technical Feature 3.D.ii.2: In this case, the second block among the two BCC blocks may include: the remaining RA field (=‘total number of RA fields required for RU allocation’ minus two)) excluding a RA field included in the first block among all RA fields (i.e., total RA fields required for RU allocation); a CRC field (e.g., 4 bits); and a Tail field (e.g., 6 bits). In other words, when the total number of RU allocation subfields required for RU allocation is four (4), the second BCC block may include two (=4 minus 2) RU allocation subfields. In addition, when the total number of RU allocation subfields required for RU allocation is eight (8), 6 (=8 minus 2) RU allocation subfields may be included in the second BCC block.

(308) Technical Feature 3.E: The above explained ‘Technical Feature 3.D’ is based on the number of total RU allocation subfields (i.e., RA fields) required for RU allocation. Since the number of total RU allocation subfields (i.e., RA fields) required for RU allocation can be determined based on the BW of the PPDU, the above-described ‘Technical Feature 3.D’ can also be expressed based on the BW of the PPDU. For example, based on the conventional 11ax standard, the total number of RU allocation subfields required within one content channel (e.g., HE-SIG content channel) for RU allocation of 40/80/160 MHz HE-PPDU was 1/2/4. Accordingly, one content channel (e.g., EHT-SIG content channel), the total number of RU allocation subfields required may be 1/2/4/8.

(309) For example, if the above ‘Technical Feature 3.D’ is explained based on the BW of the PPDU, it is the same as ‘Technical Feature 3.E’ below. Accordingly, the Technical Feature 3.E’ may include substantially the same content as the above ‘Technical Feature 3.D’.

(310) Feature 3.E.i: For example, when the bandwidth of the PPDU is 80 MHz or less, the common field of one content channel (e.g., EHT-SIG content channel) may consist of one BCC block. In this case, the one BCC block may include total RU allocation subfields (i.e., two RU allocation subfields) necessary for RU allocation. In addition, the one BCC block may include overflowed information, a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(311) Technical Feature 3.E.ii: For example, when the bandwidth of the PPDU is 160 MHz or more, as in the ‘Technical Feature 3.D.ii’, a common field of one content channel (e.g., EHT-SIG content channel) may be composed of two BCC blocks, and the specific contents may be as follows.

(312) Technical Feature 3.E.ii.1: As described above, the first block of the two BCC blocks described above includes overflowed information, two RU allocation subfields, a CRC field (e.g., 4 bits), and a Tail field (e.g., 6 bits).

(313) Technical Feature 3.E.ii.2: As discussed above, the second block among the above-discussed two BCC blocks may include: the remaining RA field (=‘total number of RA fields required for RU allocation’ minus two)) excluding a RA field included in the first block among all RA fields (i.e., total RA fields required for RU allocation); a CRC field (e.g., 4 bits); and a Tail field (e.g., 6 bits). In other words, the total number of RU allocation subfields required for the 160 MHz PPDU is four (4), and accordingly, two (=4 minus 2) RU allocation subfields may be included in the second BCC block for the 160 MHz PPDU. In addition, the total number of RU allocation subfields required for the 320 MHz PPDU is eight (8), and accordingly, six (=8 minus 2) RU allocation subfields may be included in the second BCC block for the 320 MHz PPDU.

(314) An example of the present specification described above may be expressed in various ways.

(315) FIG. 25 is a procedure flow diagram illustrating operations performed at a transmitting STA.

(316) As shown, a transmitting STA may configure a transmission (TX) physical protocol data unit (PPDU) for at least one user. (S2510) The TX PPDU may be the PPDU shown in FIG. 13 and/or FIG. 19. For example, the RX PPDU may be an EHT PPDU or a PPDU of a next version of the EHT PPDU.

(317) The TX PPDU may include a first control signal field for interpreting the TX PPDU, a second control signal field including additional control information for the TX PPDU, and a data field. The first control signal field may be a universal signal field (U-SIG) field, and the second control signal field may be an EHT-SIG field. The PPDU may include a plurality of legacy fields including the L-STF, L-LTF, L-SIG, and RL-SIG shown in FIG. 13/19, and the first control signal field may be contiguous to the plurality of legacy fields. The second control signal field may be the EHT-SIG field shown in FIGS. 20 to 24.

(318) The first control signal field may comprise a physical (PHY) version identifier including 3-bit information related to a version of the TX PPDU, a bandwidth field including 3-bit information related to the transmission bandwidth, an Uplink/Downlink (UL/DL) field including 1-bit information related to a transmission direction of the transmission PPDU, a basic service set (BSS) field including 6-bit information related to a BSS of the TX PPDU, and/or a transmission opportunity (TXOP) field including 7-bit information related to a TXOP of the TX PPDU.

(319) More specifically, the first control signal field is transmitted through two symbols, of which the first symbol (e.g., U-SIG-1 symbol) includes a total of 20 bits consisting of the above-mentioned PHY version identifier, the bandwidth field, the UL/DL field, the BSS field and the TXOP field, further includes a 6-bit reserve field capable of performing additional functions. That is, the first symbol (e.g., U-SIG-1 symbol) may be configured as 26 bits without a CRC field and a Tail field.

(320) The second symbol (e.g., U-SIG-2 symbol) of the first control signal field includes a PPDU type field including 2-bit type information, a 1-bit reserve field, a 5-bit puncturing channel information field, and a 1-bit reserve field, a 5-bit MCS field including MCS information applied to the second control signal field, and 5-bit symbol number information related to a number of symbols included in the second control signal field. Additionally, the second symbol (e.g., U-SIG-2 symbol) may consist of 26 bits, including a 4-bit CRC bit and a 6-bit Tail field. The 4-bit CRC bit of the second symbol (e.g., U-SIG-2 symbol) is configured based on: a total 26-bit of the first symbol (e.g., U-SIG-1 symbol); and a total 16-bit of the second symbol (e.g., U-SIG-2 symbol).

(321) The second control signal field may be transmitted through a plurality of content channels (e.g., EHT-SIG content channels) including first and second content channels, and the plurality of content channels may be allocated to different frequency bands. Each of the first and second content channels may have a bandwidth of 20 MHz, and each of the first and second content channels may be duplicated on frequency for the transmission bandwidth. For example, in the case of an 80 MHz PPDU, as shown in FIG. 20, a plurality of content channels may be sequentially arranged in the order of a first content channel, a second content channel, a first content channel, and a second content channel. For example, in the case of a 160 MHz PPDU, a plurality of content channels may be sequentially arranged in the order of 1, 2, 1, 2, 1, 2, 1, 2. In the case of a 160 MHz PPDU, a plurality of content channels may be sequentially arranged in the order of 1, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1, 2, 1, 2.

(322) The second control signal field may include a common field including at least one encoding block and a user specific field including at least one encoding block.

(323) The common field may include information overflowed from the first control signal field and at least one RU allocation sub-field. The overflowed information may include, for example, 4-bit information related to whether the Spatial Reuse mode can be used for the PPDU, GI of EHT-LTF included in the PPDU (indicating information related to 0.8/1.6/3.2 μ s), 2-bit information related to the LTF size (indicating information related to $2 \times$ LTF and $4 \times$ LTF), 3-bit information related to the number of EHT-LTF symbols included in the PPDU, and the like. In addition, each of the one RU allocation subfield may have a length of 9 bits and may include information related to RU allocation/location.

(324) A common field included in one content channel (e.g., EHT-SIG content channel) may be configured based on at least one encoding block (e.g., BCC encoding block). The number of

encoding blocks (e.g., BCC encoding blocks) included in one common field included in one content channel (e.g., EHT-SIG content channel) may be set in various ways based on the bandwidth of the PPDU. Also, information elements included in at least one encoding block may be set in various ways based on the bandwidth of the PPDU.

(325) For example, when a 20/40/80 MHz PPDU is transmitted/received, one common field included in one content channel (e.g., EHT-SIG content channel) can be configured based on only one encoding block. One corresponding encoding block may include the overflowed information described above and one or two RU allocation subfields. In this case, the number of RU allocation subfields may be set to two (2) for the 80 MHz PPDU and one (1) for the 20/40 MHz PPDU. The one encoding block includes a 4-bit CRC field, and the CRC field may be generated based on the overflowed information and one or two RU allocation subfields. One encoding block may include a 6-bit tail field. The above example can be embodied in the content channel of FIG. 28.

(326) FIG. 28 is an example of a content channel for a 20/40/80 MHz PPDU. An example of FIG. 28 shows any one of a plurality of content channels transmitting the second control signal field. The example of FIG. 28 is an example in which a user-specific field is specifically added to the examples of FIGS. 20 to 24.

(327) For example, when a 160 MHz PPDU is transmitted/received, one common field included in one content channel (e.g., EHT-SIG content channel) may be configured based on two contiguous encoding blocks. Among the two encoding blocks, a first block may consist of the overflowed information described above, two RU allocation subfields, a 4-bit CRC field, and a 6-bit tail field. Among the two encoding blocks, the second block may consist of two RU allocation subfields, a 4-bit CRC field, and a 6-bit tail field. The above example can be embodied in the content channel of FIG. 29.

(328) FIG. 29 is an example of a content channel for a 160 MHz PPDU. An example of FIG. 29 shows any one of a plurality of content channels transmitting the second control signal field. The example of FIG. 29 is an example in which a user-specific field is specifically added to the examples of FIGS. 20 to 24.

(329) For example, when a 320 MHz PPDU is transmitted/received, one common field included in one content channel (e.g., EHT-SIG content channel) may be configured based on two contiguous encoding blocks. Among the two encoding blocks, a first block may include the overflowed information described above, two RU allocation subfields, a 4-bit CRC field, and a 6-bit tail field. Among the two encoding blocks, the second block may consist of 6 RU allocation subfields, a 4-bit CRC field, and a 6-bit tail field. The above-described example may be embodied in the content channel of FIG. 30.

(330) FIG. 30 is an example of a content channel for a 320 MHz PPDU. An example of FIG. 30 shows any one of a plurality of content channels transmitting the second control signal field. The example of FIG. 30 is an example in which a user-specific field is specifically added to the examples of FIGS. 20 to 24.

(331) The overflowed information described above may be equally/identically included in all content channels as described in FIG. 20 and the like.

(332) As shown, the TX STA may transmit the above-described TX PPDU to a reception (RX) STA. (S2520)

(333) FIG. 26 is another procedure flow diagram illustrating operations performed at a TX STA.

(334) Step S2510 of FIG. 25 may be expressed in detail as explained in steps S2610, S2620, and S2630 of FIG. 26. Steps S2610, S2620, and S2630 do not necessarily need to be performed sequentially and may be performed simultaneously or in a manner different from the illustrated order.

(335) The TX STA may determine whether the bandwidth of the transmitting PPDU to be configured is greater than or equal to 160 MHz, and less than or equal to 80 MHz. For example, the number of encoding blocks included in the common field transmitted through one content channel

is set to one (1) for the TX PPDU having a transmission bandwidth of 80 MHz or less, and is set to two (2) for the RX PPDU having a transmission bandwidth of 160 MHz or more.

(336) For example, if the bandwidth of the transmitted PPDU is 160 MHz or more, the PPDU may be configured based on the first bandwidth (i.e., 160 MHz or 320 MHz bandwidth). (S2620). That is, the PPDU may be configured based on the first bandwidth (i.e., 160 MHz or 320 MHz bandwidth) in the same manner as in step S2510 described above.

(337) For example, when the bandwidth of the transmitted PPDU is 80 MHz or less, the PPDU may be configured based on the second bandwidth (i.e., 20 MHz, 40 MHz, or 80 MHz bandwidth). (S2620). That is, the PPDU may be configured based on the second bandwidth (i.e., 20 MHz, 40 MHz or 80 MHz bandwidth) in the same manner as in step S2510 described above. (S2620).

(338) The operation of FIGS. 25 and/or 26 may be performed by the apparatus of FIGS. 1 and/or 14. For example, the TX STA may be implemented as the apparatus of FIGS. 1 and/or 14. The processor of FIGS. 1 and/or 14 may perform the operations of FIGS. 25 and/or 26 described above. In addition, the transceiver of FIGS. 1 and/or 14 may perform the transmission/reception operations described in FIGS. 25/26.

(339) In addition, the apparatus proposed in this specification does not necessarily include a transceiver, and may be implemented in a chip form including a processor and a memory. Such an apparatus may generate/store a TX PPDU according to the example described apparatus. Such an apparatus may be connected to a separately manufactured transceiver to support actual transmission and reception.

(340) FIG. 27 is a procedure flow diagram illustrating operations performed in a reception (RX) STA.

(341) As shown, the RX STA may receive a reception (RX) physical protocol data unit (PPDU). (S2710). Specific technical characteristics of the RX PPDU are the same as those of the TX PPDU described in this specification, and may be the same as those of the PPDU described in FIGS. 25/26.

(342) As shown, the RX STA may decode the RX physical protocol data unit (PPDU) based on the first control signal field and the second control signal field. (S2710). For example, as described above, the first control signal field includes various pieces of information about the version of the PPDU, the bandwidth of the PPDU, the type of the PPDU, and the second control signal field. The receiving STA may start decoding the RX PPDU based on the information of the first control signal field. In addition, the receiving STA decodes the second control signal field based on various information (e.g., allocation information about RU, MCS information about data field, etc.) included in the second control signal field, and based on this, user data included in the data field can be decoded.

(343) This specification proposes a computer readable medium (CRM) implemented in various forms. A computer readable medium according to the present specification may be encoded as at least one computer program including instructions. Instructions stored in the medium may control the processor described in FIG. 1 and/or FIG. 14 and the like. That is, the instructions stored in the medium control the processor presented in this specification to perform the above-described operations of the TX STA and the RX STA (e.g., FIG. 25 to FIG. 27).

(344) The foregoing technical features of the present specification are applicable to various applications or business models. For example, the foregoing technical features may be applied for wireless communication of a device supporting artificial intelligence (AI).

(345) Artificial intelligence refers to a field of study on artificial intelligence or methodologies for creating artificial intelligence, and machine learning refers to a field of study on methodologies for defining and solving various issues in the area of artificial intelligence. Machine learning is also defined as an algorithm for improving the performance of an operation through steady experiences of the operation.

(346) An artificial neural network (ANN) is a model used in machine learning and may refer to an

overall problem-solving model that includes artificial neurons (nodes) forming a network by combining synapses. The artificial neural network may be defined by a pattern of connection between neurons of different layers, a learning process of updating a model parameter, and an activation function generating an output value.

(347) The artificial neural network may include an input layer, an output layer, and optionally one or more hidden layers. Each layer includes one or more neurons, and the artificial neural network may include synapses that connect neurons. In the artificial neural network, each neuron may output a function value of an activation function of input signals input through a synapse, weights, and deviations.

(348) A model parameter refers to a parameter determined through learning and includes a weight of synapse connection and a deviation of a neuron. A hyper-parameter refers to a parameter to be set before learning in a machine learning algorithm and includes a learning rate, the number of iterations, a mini-batch size, and an initialization function.

(349) Learning an artificial neural network may be intended to determine a model parameter for minimizing a loss function. The loss function may be used as an index for determining an optimal model parameter in a process of learning the artificial neural network.

(350) Machine learning may be classified into supervised learning, unsupervised learning, and reinforcement learning.

(351) Supervised learning refers to a method of training an artificial neural network with a label given for training data, wherein the label may indicate a correct answer (or result value) that the artificial neural network needs to infer when the training data is input to the artificial neural network. Unsupervised learning may refer to a method of training an artificial neural network without a label given for training data. Reinforcement learning may refer to a training method for training an agent defined in an environment to choose an action or a sequence of actions to maximize a cumulative reward in each state.

(352) Machine learning implemented with a deep neural network (DNN) including a plurality of hidden layers among artificial neural networks is referred to as deep learning, and deep learning is part of machine learning. Hereinafter, machine learning is construed as including deep learning.

(353) The foregoing technical features may be applied to wireless communication of a robot.

(354) Robots may refer to machinery that automatically process or operate a given task with own ability thereof. In particular, a robot having a function of recognizing an environment and autonomously making a judgment to perform an operation may be referred to as an intelligent robot.

(355) Robots may be classified into industrial, medical, household, military robots and the like according uses or fields. A robot may include an actuator or a driver including a motor to perform various physical operations, such as moving a robot joint. In addition, a movable robot may include a wheel, a brake, a propeller, and the like in a driver to run on the ground or fly in the air through the driver.

(356) The foregoing technical features may be applied to a device supporting extended reality.

(357) Extended reality collectively refers to virtual reality (VR), augmented reality (AR), and mixed reality (MR). VR technology is a computer graphic technology of providing a real-world object and background only in a CG image, AR technology is a computer graphic technology of providing a virtual CG image on a real object image, and MR technology is a computer graphic technology of providing virtual objects mixed and combined with the real world.

(358) MR technology is similar to AR technology in that a real object and a virtual object are displayed together. However, a virtual object is used as a supplement to a real object in AR technology, whereas a virtual object and a real object are used as equal statuses in MR technology.

(359) XR technology may be applied to a head-mount display (HMD), a head-up display (HUD), a mobile phone, a tablet PC, a laptop computer, a desktop computer, a TV, digital signage, and the like. A device to which XR technology is applied may be referred to as an XR device.

(360) FIG. 31 is an example of a common field and a user specific field of the present specification. As shown in FIG. 31, a common field related to an RU allocation subfield and a user specific field related to user fields for RUs signaled in the common field can be defined for the present specification.

Claims

1. A method in a wireless Local Area Network, the method comprising: configuring a physical protocol data unit (PPDU) including a universal signal (U-SIG) field and an extremely high throughput signal (EHT-SIG) field, wherein the EHT-SIG field includes a first content channel and a second content channel, wherein one content channel includes a common field comprising at least one two encoding blocks and a user specific field comprising encoding block, wherein a first encoding block of the common field includes U-SIG overflow information, two resource unit (RU) allocation subfields, a 4-bit cyclic redundancy code (CRC) field, and a 6-bit Tail field, wherein a second encoding block of the common field includes two RU allocation subfields, a 4-bit CRC field, and a 6-bit Tail field, wherein the first content channel allocated on a first frequency band includes a first common field including the U-SIG overflow information and RU allocation subfields related to first tone ranges, wherein the second content channel allocated on a second frequency band includes a second common field including the U-SIG overflow information and RU allocation subfields related to second tone ranges, wherein the first content channel is duplicated on a third frequency band, wherein the second content channel is duplicated on a fourth frequency band, wherein the first to fourth frequency bands are sequentially allocated within an 80 MHz frequency subblock, wherein the first tone ranges comprise a tone range of $\{-500:-259\}$, and the second tone ranges comprise a tone range of $\{259:500\}$; and transmitting the PPDU.
2. The method of claim 1, wherein each of the two RU allocation subfields has a length of 9 bits.
3. The method of claim 1, wherein a bandwidth of the PPDU is 160 MHz.
4. The method of claim 1, wherein the PPDU further includes a legacy signal (L-SIG) field, and a repeated legacy signal (RL-SIG) field which is a repeat of the L-SIG field and is contiguous to the L-SIG field, wherein the U-SIG field is contiguous to the RL-SIG field, wherein the L-SIG field includes a length field, the length field has a value satisfying condition that a remainder is zero when the length field is divided by three, and the remainder is used to differentiate the PPDU from a High Efficiency (HE) PPDU.
5. The method of claim 4, wherein the RL-SIG field includes four extra subcarriers related to channel estimation at indices of $\{-28, -27, 27, 28\}$, and values of the extra subcarriers are $\{-1, -1, -1, 1\}$, respectively.
6. The method of claim 5, wherein the U-SIG field has a length of 2 symbols, wherein the U-SIG field includes version-independent bits followed by version-dependent bits, wherein the version-independent bits are configured based on five different subfields which are a physical (PHY) version identifier subfield having a length of 3 bits, a uplink/downlink subfield having a length of 1 bit, a Basic Service Set (BSS) color subfield, a transmission opportunity (TXOP) subfield, and a bandwidth subfield.
7. A station (STA) in a wireless Local Area Network, the STA comprising: at least one processor; and at least one computer memory operatively connected to the at least one processor and storing instructions that, based on being executed by the at least one processor, perform operations comprising: receiving a physical protocol data unit (PPDU) including a universal signal (U-SIG) field and an extremely high throughput signal (EHT-SIG) field, wherein the EHT-SIG field includes a first content channel and a second content channel, wherein one content channel includes a common field comprising two encoding blocks and a user specific field comprising at least one encoding block, wherein a first encoding block of the common field includes U-SIG overflow information, two resource unit (RU) allocation subfields, a 4-bit cyclic redundancy code

(CRC) field, and a 6-bit Tail field, wherein a second encoding block of the common field includes two RU allocation subfields, a 4-bit CRC field, and a 6-bit Tail field, wherein the first content channel allocated on a first frequency band includes a first common field including the U-SIG overflow information and RU allocation subfields related to first tone ranges, wherein the second content channel allocated on a second frequency band includes a second common field including the U-SIG overflow information and RU allocation subfields related to second tone ranges, wherein the first content channel is duplicated on a third frequency band, wherein the second content channel is duplicated on a fourth frequency band, wherein the first to fourth frequency bands are sequentially allocated within an 80 MHz frequency subblock, wherein the first tone ranges comprise a tone range of $\{-500:-259\}$, and the second tone ranges comprise a tone range of $\{259:500\}$.

8. The STA of claim 7, wherein each of the two RU allocation subfields has a length of 9 bits.

9. The STA of claim 7, wherein a bandwidth of the PPDU is 160 MHz.

10. The STA of claim 7, wherein the PPDU further includes a legacy signal (L-SIG) field, and a repeated legacy signal (RL-SIG) field which is a repeat of the L-SIG field and is contiguous to the L-SIG field, wherein the U-SIG field is contiguous to the RL-SIG field, wherein the L-SIG field includes a length field, the length field has a value satisfying condition that a remainder is zero when the length field is divided by three, and the remainder is used to differentiate the PPDU from a High Efficiency (HE) PPDU.

11. The STA of claim 10, wherein the RL-SIG field includes four extra subcarriers related to channel estimation at indices of $\{-28, -27, 27, 28\}$, and values of the extra subcarriers are $\{-1, -1, -1, 1\}$, respectively.

12. The STA of claim 11, wherein the U-SIG field has a length of 2 symbols, wherein the U-SIG field includes version-independent bits followed by version-dependent bits, wherein the version-independent bits are configured based on five different subfields which are a physical (PHY) version identifier subfield having a length of 3 bits, a uplink/downlink subfield having a length of 1 bit, a Basic Service Set (BSS) color subfield, a transmission opportunity (TXOP) subfield, and a bandwidth subfield.

13. A method in a wireless Local Area Network, the method comprising: receiving a physical protocol data unit (PPDU) including a universal signal (U-SIG) field and an extremely high throughput signal (EHT-SIG) field, wherein the EHT-SIG field includes a first content channel and a second content channel, wherein one content channel includes a common field comprising two encoding blocks and a user specific field comprising at least one encoding block, wherein a first encoding block of the common field includes U-SIG overflow information, two resource unit (RU) allocation subfields, a 4-bit cyclic redundancy code (CRC) field, and a 6-bit Tail field, wherein a second encoding block of the common field includes two RU allocation subfields, a 4-bit CRC field, and a 6-bit Tail field, wherein the first content channel allocated on a first frequency band includes a first common field including the U-SIG overflow information and RU allocation subfields related to first tone ranges, wherein the second content channel allocated on a second frequency band includes a second common field including the U-SIG overflow information and RU allocation subfields related to second tone ranges, wherein the first content channel is duplicated on a third frequency band, wherein the second content channel is duplicated on a fourth frequency band, wherein the first to fourth frequency bands are sequentially allocated within an 80 MHz frequency subblock, wherein the first tone ranges comprise a tone range of $\{-500:-259\}$, and the second tone ranges comprise a tone range of $\{259:500\}$; and decoding the PPDU based on the U-SIG field and the EHT-SIG field.

14. The method of claim 13, wherein each of the two RU allocation subfields has a length of 9 bits.

15. The method of claim 13, wherein a bandwidth of the PPDU is 160 MHz.

16. The method of claim 13, wherein the PPDU further includes a legacy signal (L-SIG) field, and a repeated legacy signal (RL-SIG) field which is a repeat of the L-SIG field and is contiguous to

the L-SIG field, wherein the U-SIG field is contiguous to the RL-SIG field, wherein the L-SIG field includes a length field, the length field has a value satisfying condition that a remainder is zero when the length field is divided by three, and the remainder is used to differentiate the PPDU from a High Efficiency (HE) PPDU.

17. The method of claim 16, wherein the RL-SIG field includes four extra subcarriers related to channel estimation at indices of $\{-28, -27, 27, 28\}$, and values of the extra subcarriers are $\{-1, -1, 1, 1\}$, respectively.

18. The method of claim 17, wherein the U-SIG field has a length of 2 symbols, wherein the U-SIG field includes version-independent bits followed by version-dependent bits, wherein the version-independent bits are configured based on five different subfields which are a physical (PHY) version identifier subfield having a length of 3 bits, a uplink/downlink subfield having a length of 1 bit, a Basic Service Set (BSS) color subfield, a transmission opportunity (TXOP) subfield, and a bandwidth subfield.
